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ICAR

NAARM

National Academy of Agricultural Research Management



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1991 - 92



**NATIONAL ACADEMY OF
AGRICULTURAL RESEARCH MANAGEMENT
Rajendranagar, Hyderabad - 500 030 (India)**

<i>Editor-in-Chief</i>	:	T.Balaguru
<i>Editor</i>	:	R.V.V.S. Prakasa Rao
<i>Editorial Assistance</i>	:	Vijay Kumar
<i>Photography</i>	:	L. Venkateswarlu
<i>Cover Design</i>	:	P. Namdev

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INTRODUCTION

Historical Background

It is now well recognized that management holds the key to increased productivity in research, as it does to production and productivity in several other sectors. Management principles are being increasingly applied in agricultural research, education and extension. The need for an institutional set-up to provide training on principles of scientific management of agricultural research was felt two-and-a-half decades ago by the Indian Council of Agricultural Research (ICAR) and the Government of India, following the organization of a training course in Agricultural Project Management, in response to a request from the FAO. A National Training Centre for Agriculture Development and Project Planning functioned at Indian Agricultural Research Institute (IARI), New Delhi during the Fourth Five Year Plan. This further strengthened the need to continue these training activities, and to establish an organisation for this purpose.

A working group, set up during the Fifth Five Year Plan, recommended the establishment of a Central Staff College for Agriculture to function as national and international repository of ideas and information in the field of agricultural research and development. The proposed College was also required to act as a clearing house for dissemination and utilisation of Research Management information, and also to undertake a programme of publishing high quality training material and allied publications considered important for the priority areas of education.

After more detailed deliberations and planning, the Central Staff College for Agriculture came into existence on September 1, 1976. This was later renamed as the National Academy of Agricultural Research Management from 1979.

The Academy is located at the campus of the Andhra Pradesh Agricultural University (APAU) on 50 hectares of land provided by the University to the Academy on a long-term lease. The NAARM campus is situated at Rajendranagar, a suburb of Hyderabad, about 15 km south of the city.

Several functional buildings have now been constructed in this campus. These include an Administrative Block, Faculty-cum-Laboratory Building, an Academic Block which has several class and discussion rooms, Halls of Residence and Scientists Home. Besides, residential quarters for all categories of staff have also been constructed. Several instructional and technology demonstration facilities have also been developed. An Advanced Centre on Agricultural Education and Research Management has also been established with the support of UNDP/UNESCO/FAO. These physical and infrastructural facilities as well as the technical strength of the programmes have placed the Academy as an important National and International Centre serving in the country and the South and Southeast Asian Region.

OBJECTIVES

The National Academy of Agricultural Research Management (NAARM) was established with the main objective of familiarizing all those concerned with the agricultural research, education and development with the principles of scientific management. It had taken up the responsibility to plan, organize and conduct training programmes, workshops and seminars in various areas related to agricultural research management for the scientists of ICAR and Agricultural Universities. It is functioning as a repository of ideas and information, both at national and international levels, and is fulfilling its role as a clearing house for dissemination and utilization of such information and know-how. The Academy had also started training programmes in the areas of educational technology, curriculum development and structured learning. Research programmes and case studies in various fields of agricultural research management and educational technology were also taken up.

The main objectives of the Academy are :

- * to organize and conduct training programmes in agricultural research management to scientists at various levels,
- * to undertake systematic review and study of management problems of ICAR Institutes, State Agricultural Universities (SAUs), Agricultural Research Projects and Systems,
- * to organize, liaise and coordinate programmes of international cooperation in the field of agricultural research management,
- * to plan, organize and conduct training programmes, workshops and seminars in educational technology and structured learning for the faculty of SAUs and ICAR Institutes, and
- * to assist the Agricultural Universities in developing Regional Centres of Management to cater to the needs of the various SAUs.

HIGHLIGHTS

- * One of the hallmark of the year was the organization of an International Workshop on Implications of Farming Systems Research/Extension (FSR/E) in the Indian Context in collaboration with the Ford Foundation.
- * A National Seminar on Policy Issues in Generation, Reliability and Retrieval of Database on Indian Agricultural Sector for Social Sciences Research was organised to take stock of the availability, retrievability and reliability of different types of data pertaining to crop, livestock, fisheries and forestry sub-sectors of the agricultural sector has been felt in view of the newer thrusts and directions in which agro-economic research, as a base for policy planning and development, is moving.
- * A 2-day National Consultation Meeting on ICAR Personnel Policies was conducted at the Academy in which working scientists and policy makers discussed several policy issues and came out with many useful suggestions for improvement.
- * Many new programmes were started which include:
 - i) Course on Computer Applications in Agricultural Research and Education for the ICAR scientists to expose them to various computer packages like dBase, Reflex, Word Star, M. Stat, Lotus, etc.; and ii) Course on Training of Trainers for the ICAR scientists and Agricultural University teachers to expose them to recent trends in training with a view to improving their effectiveness as trainers.
- * A short-term, specialized training programme was organized for the scientists from Bangladesh, who attended three senior level programmes, viz. Course on Agricultural Research Project Management, Course on Human Resource Management, and Programme on Administrative and Financial Management. Besides institutional visits were arranged in three phases to familiarize them with the Indian Agricultural Research and Education Systems.

- * The General Information Service Terminal National Informatics Centre (GISTNIC), a dial-for-data service, was installed at the Academy for the benefit of the trainees and the faculty members. Besides general information about Indian agriculture and communication facilities, it offers a few other databases in specialised areas for use by the administrators, economists and financial analysts.
- * Dr A. Gopalam and Dr M.M. Anwer, both Associate Professors, underwent advanced training in USA for a period of four-and-a-half months under the auspices of the UNDP Advanced Centre on Agricultural Education and Research Management.
- * A three member delegation from People's Republic of China comprising Dr Chen Wanjin, Vice President, Dr Zhang Lijian, Deputy Director of International Exchange Division and Mr Wang Zhongqiy, Senior Executives from the Chinese Academy of Agricultural Sciences, visited the Academy.

TRAINING PROGRAMMES

The Academy organizes several training programmes to fulfil the needs of the clientele groups at various levels of management in the agricultural research and development organisations. Three levels of clientele - junior, middle and senior - are identified for training purposes.

The list of training programmes and workshops organized during the year is given below:

Sl. No.	Title of the Programme	Dates	No. of Participants
<i>Junior level:</i>			
1.	36th Foundation Course	15.04.91 to 13.09.91	23
2.	37th Foundation Course	11.06.91 to 08.11.91	57
3.	38th Foundation Course	05.08.91 to 04.01.92	82
4.	39th Foundation Course	03.12.91 to 25.04.92	37
<i>Midale and Senior levels:</i>			
1.	Workshop on Meaningful Learning as a Communication Process	09.06.91 to 12.06.91	16
2.	Workshop on Management Information System for Agricultural Research	10.06.91 to 20.06.91	12
3.	Workshop on Bibliometric and Scientometric Tools for the Management of Productivity in Agricultural Research	25.06.91 to 29.06.91	33
4.	Course on Computer Applications in Agricultural Research	16.07.91 to 23.07.91	23
5.	Course on Agricultural Research Project Management	05.08.91 to 17.08.91	26
6.	Course on Management of Human Resources	20.08.91 to 31.08.91	26

7.	Programme on Administrative and Financial Management	02.09.91 to 06.09.91	14
8.	Workshop on Management Information System for Agricultural Research	03.09.91 to 13.09.91	13
9.	Course on Curriculum Design and Development	17.09.91 to 28.09.91	17
10.	Course on Agricultural Research Project Management	08.10.91 to 19.10.91	28
11.	Course on Training of Trainers	15.10.91 to 26.10.91	29
12.	Two day National Consultation on ICAR Personnel Policies	22.10.91 to 24.10.91	40
13.	Workshop on Monitoring and Evaluation of Agricultural Research Projects	22.10.91 to 24.10.91	23
14.	Workshop on Effective Instructional Methods	29.10.91 to 02.11.91	15
15.	Orientation Course on Fisheries Extension Education	08.11.91 to 23.11.91	16
16.	National Seminar on Policy Issues in Generation, Flexibility and Retrieval of Database in Indian Agricultural Sector for Social Sciences Research	13.11.91 to 15.11.91	27
17.	International Workshop on Implications of Farming Systems Research/Extension (FSR/E) in the Indian Context	25.11.91 to 28.11.91	54
18.	Course on Information Management and Research Communication in Agriculture	26.11.91 to 06.12.91	21
19.	Course on Management of people at Work	10.12.91 to 21.12.91	26
20.	Workshop on Motivation Management for Farmers and Community Participation	20.01.92 to 23.01.92	31

Course for Junior Scientists

A regular course is offered to the scientists recruited to the Agricultural Research Service (ARS) by the ICAR. The training programme, in operation since 1976, had undergone structural changes in concept, duration and details of operation during this period. The experience of the last 14 years and the feedback obtained from the participants have been considered in modifying and redesigning the programme to meet the emerging requirements. In the current approach, emphasis is laid on pro-active learning and integrated experiential learning, so as to build skills in competency, analysis, synthesis, application and formulation by providing hands-on experience to enable the learner adopt a holistic approach and systems thinking in identifying and managing projects. The course is organised within the framework of the following objectives:

- * to acquaint the scientists with the general agricultural scenario in the country with reference to research and development,
- * to equip them with techniques in research project management,
- * to give an insight into the rural society and structure,
- * to develop skills in organisational behaviour for high motivation, team work and conflict resolution,
- * to provide training in agricultural communication and transfer of technology,
- * to orient the ARS scientists to the ICAR and the other National Agricultural Research, Education and Extension Education Subsystems in the country, and
- * to foster a feeling of fraternity among the members of the ARS serving in the different parts of the country.

In order to fulfil the above objectives, the course has been structured in three phases with theoretical and practical classes.

In Phase I, spent at the Academy, instruction on various aspects of project management and national priorities in agriculture are given emphasis, after a detailed exposure to the National Agricultural Research, Education and Extension Systems in India. During Phase II, the trainees are attached in small groups to the Regional Research Stations of the Agricultural Universities or ICAR Institutes where they receive practical training in project identification and formulation, technology development process and mode of transfer of technology, besides developing an empathy for farmers and appreciation for rural life. Phase III, spent at the Academy, is the follow-up and refinement of the experiences for conceptual development in all related areas like agricultural communication, audio-visual aids, treatment of research findings and presentation of scientific papers. During the period under report, 199 ARS scientists freshly recruited into the service completed their training.

Courses for Middle and Senior Level Clientele

Programme on Administrative and Financial Management (Coordinator : V. Raghavendra Rao)

The Management Development Programme was organized for Senior Executives of ICAR including Deputy Directors General and Assistant Director Generals at headquarters and Directors of ICAR Institutes. This programme is designed to improve the managerial effectiveness of senior executives of ICAR in the fields of 'Administrative and Financial Management' by giving them exposure to the concepts in the above areas with reference to management of agricultural research. The course was organized from September 2 to 6, 1991.

The programme covered the following topics:

Administrative efficiency and productivity; Motivating the personnel related to administration and financial management; Computerization of office procedures; Authority and accountability; Legal issues; Labour management; Performance appraisal; Communication conflicts; Disciplinary issues; Vigilance issues; Scope of finance function in research institutes; Project and budget formulation in ICAR institutes; Budget and resources allocation at ICAR; Approaches to budgeting - A zero base orientation; Project based monitoring at ICAR institutes; Performance monitoring at ICAR; MIS for Financial planning and monitoring; Panel discussions - Implementing improvements in Systems i) Administrative management, ii) Financial management.

Fourteen participants including four Senior Executives from Bangladesh and Assistant Director Generals from ICAR headquarters and Directors of ICAR Institutes attended the programme.

Course on Agricultural Research Project Management (CARPM) (Coordinators : K. Venkateswarlu & S.N. Saha)

The course is designed to provide exposure to the research managers in the ICAR Institutes to the concepts of research projects and application of 'management approach' while dealing with the problems faced by the agricultural research units. It is offered to the senior research managers, who heading the divisions in the ICAR Institutes in the cadre of Principal Scientist, and Professors and Heads of Departments in the SAUs. The duration of the course is two weeks. This course was offered twice during August 5- 17, 1991 and October 8 - 19, 1991. Twenty six and 28 participants attended the courses respectively

The topics covered in the course include:

Project Planning - Importance of planning long-term and short-term projects in the management process and systematic project planning; Project selection - Methods and criteria used for project selection; Monitoring and evaluation - Network methods for time scheduling, control and monitoring and resource smoothing of research projects; Organisation structure - Principles of organisation and the importance of technology and environment in organisation design; Group dynamics - Characteristics and classification of groups, group decision-making and inter-group problems in organisations; Leadership - Role of leadership in management and leadership styles suitable to Indian conditions; Conflicts - Types of conflicts that occur in organisation and the methods used to resolve the conflicts.

Course on Management of Human Resources
(Coordinators: M.M. Anwer & Jagannadham Challa)

The Course is designed to expose the Research Managers to the concepts of human behaviour in organisation and topics related to human resources management in agricultural research. This course is organized for senior scientists, who are heading divisions in the ICAR Institutes in the cadres of Principal Scientist, and Professors and Heads of Departments in the SAUs. The duration of the course is two weeks. The course was organized once during August 20 - 31, 1991.

Twenty six participants attended the course, which covered the following topics:

Individual and the organisation; Motivation of the scientists; Interpersonal relationships; Transactional Analysis; Group dynamics; Leadership; Teamwork; Conflicts; Personnel policies; Structure and design of organisation; Organisational communication; Mass communication; Research project management.

Course on Computer Applications in Agricultural Research and Education

(Coordinators: K. Vidyasagar Rao & U. Muralidhar)

This Course was organized from July 16 to 27, 1991 to expose the participants to computer uses for learning and teaching and were also exposed to different packages like CDS, dBase, Reflex, Word Star, M. Stat, Lotus, etc. In addition, the following lectures were also covered:

Basics of computers, Computer application in agricultural research and Database management system.

Twenty three participants from different ICAR Institutes attended the course. Some of the participants expressed the desire to have an advanced programme on computers and some of them wanted the introduction of packages like Word Perfect, SPSS in the course.

Course on Curriculum Design and Development (Coordinator: Devesh Kishore)

The Course was organized from September 17 to 28, 1992. It was designed to expose the participants to the philosophy, principles and processes related to curriculum development and modification at macro and micro levels keeping in view the students coming from various education streams, changing needs of the society and expectations of the students and employers, to enable participants understand the process of developing effective instructional systems and design. An opportunity was also provided to them to prepare suitable curriculum at micro levels using the latest methodology, and to develop guidelines for the teachers and the students to meet such requirements.

The course contents include Educational technology - Its role, process and guiding principles for curriculum development; Curriculum design and development - Model approaches in curriculum at macro and micro levels; Instructional system -- Designs and methods; Scientific information management and personal documentation for agricultural scientists and teachers.

Seventeen participants from different Agricultural Universities attended the course.

Course on Training of Trainers
(Coordinators: T. Balaguru & P. Manikandan)

Training function is rapidly expanding in the ICAR Institutes and State Agricultural Universities. In order to provide an opportunity for the trainers to come together, share ideas and experiences, and become aware of the latest developments in the area of training, this specialized course was organized from October 15 to 26, 1991. The major objectives of the course were: i) to learn and use basic principles of adult education and experiential learning cycle; ii) to impart skills in designing appropriate training programmes to suit specific needs; iii) to develop skills in preparing and carrying out teaching/training sessions; iv) to use different teaching/training techniques such as self-instructional methods (audio and video tutorials, computer assisted instruction), case study, role play, group discussions, experiential lecturette, co-training, etc.; v) to improve communication and facilitation skills such as questioning, paraphrasing, summarizing, listening and feedback; and vi) to adapt ideas on various instructional and training methods for use in different situations.

A series of practice sessions were arranged for the participants. Some of the activities taken up for these sessions include: Group task accomplishment; Case study; Role play; Experiential lecturette, Co-training; and Brain storming. In addition, a variety of instruments were administered to the participants, which gave them an opportunity to identify themselves and formulate action plans for self-improvement. Some of the inventories used include: Training styles inventory; Training needs assessment tool; Communication and listening skill inventories; and Consulting style inventory. The participants were also given some guidelines for the development and implementation of effective training programme.

Twenty nine teachers/trainers with different background from the ICAR Institutes and State Agricultural Universities participated in the course. Sixteen of them came from the crop sciences; seven from animal sciences; and six from fishery sciences. In addition, three training managers from Nigeria joined the course in the second week.

**Course on Information Management and Research
Communication in Agriculture**
(Coordinator: R.B. Gaddagimath)

The Course was organized from November 26 to December 6, 1991. It aimed at organizing information and effective communication of research results. The objectives of the course were: to study issues related to project maintenance, research coordination and networks; to know computer applications for project monitoring and information management; to provide hands-on experience in technical information management process and management information systems; to identify appropriate forms of technical communication; and to share experience relating to publication policies and editorial practices of the ICAR.

The course was attended by 21 Information Officers, Technical Editors of whom 13 were from ICAR and eight were from State Agricultural Universities.

Course on Management of People at Work
(Coordinators: Jagannadham Challa & S. Shanmugam)

The Course was organized from December 10 to 21, 1991. It was designed such that participants develop an awareness within the given organisational structures, systems, policies and development process; the managers build an atmosphere of cordial transactions and interpersonal relationship within their research divisions, create an environment to motivate scientists and by developing skills of appropriate leadership styles as per the situations to lead research groups towards excellence and organizational goals through personal effectiveness, effective group dynamics and playing a significant role in various positions in the organisations and also be able to manage differences and job stress of various categories of scientists, technicians and administrative staff while managing people at work successfully, efficiently and effectively. The course covered the following topics, viz. Conceptualization, Experiential learning, Current awareness, etc.

Twenty seven Principal Scientists, Professors, Heads of the Departments from 13 ICAR Institutes and three Agricultural Universities.

Short-Term Training Programme for Participants from Bangladesh

On the request of the Principal Training Officer from Bangladesh Agricultural Research Council, Dhaka in Bangladesh, a training programme for the scientists from Bangladesh was organized, which included three courses of the Academy, viz. Course on Agricultural Research Project Management - Course on Human Resource Management, and Programme on Administrative and Financial Management. The programme also included three phases of visits for the exposure to Agricultural Research and Education System in the country.

Four participants after completing the training programmes in the Academy visited other ICAR Institutes like IIHR, Bangalore, CTRL, Bombay, IIM, Ahmedabad and IARI, New Delhi in the first phase; and the Institutes in Hyderabad viz. MANAGE, NIRD, APAU, ICRISAT in the second phase and JTRL & CIJAF, Calcutta the third phase.

The learning experiences of these participants were of great value, since the developing system of Bangladesh Agricultural Research and Education can be greatly strengthened by the input of these trainees.

Workshops

Management Information Systems for Agricultural Research (Coordinators: Jagdeesh C. Kalla & T. Balaguru)

The activities of Agricultural research institutes in the country have grown in size and scope. Currently, research managers are facing the problem of matching the available resources in terms of personnel, operating expenses and physical facilities with the requirements to meet emerging needs of sophisticated and more complex research programmes. Since appropriate information is an essential prerequisite for right decision-making, efficiently in the use of such scarce resources can be achieved only on the basis of precise information on the research projects in operation, the manpower engaged, the facilities being used and the costs involved. Lack of such information in a readily usable form and on time can be a major impediment for the research managers to effectively plan and manage research in their institutes. It becomes, therefore, necessary to develop an organized Management Information System (MIS) that can generate relevant information for the research managers to make sound decisions.

Realizing the need to help the agricultural research institutes in the country develop an organized MIS to ensure rationale-based decisions, the ICAR has recently taken a decision to launch a programme on MIS. The Academy is assisting the ICAR in this cause by training to develop the required manpower to effectively manage the MIS at various institutes. In this direction, two 10-day workshops on 'Management Information Systems for Agricultural Research, one in June from 10th to 20th and another in September from 3rd to 13th, were organized at NAARM. These workshops were organized with the following objectives:

- i) to familiarize researchers with the concept of MIS;
- ii) to help research institutes develop and use an organized MIS for effective utilization of resources like personnel, finance and facilities; and
- iii) to enable researchers develop project budgeting system (PBS) towards better research management in agriculture.

These two workshops were specially organized for the ICAR institutes. The first workshop was attended by 12 while the second was attended by 13 participants from these institutes. The workshops were built around three major themes, viz. i) Guidelines on MIS/PBS methodology; ii) Development and use of MIS/PBS; and iii) Project work based on a case study. A variety of learning techniques like experimental lecturette, computer-based exercises, case studies and group activities were employed. Efforts were made to provide the participants with considerable hands-on experience in the development and use of MIS/PBS using personal computers.

The guidelines on MIS/PBS prepared by NAARM were presented and discussed in the classroom. This was followed by intensive laboratory work where the participants practiced the development of MIS/PBS. This included the generation of various types of output with the help of software REFLEX by utilizing the real life data pertaining to various ICAR Institutes. Utilizing the data provided to them in the form of a Case study, the participants undertook a project work to develop MIS/PBS for one of the ICAR Institutes. Follow-up action plans highlighting the schedule of activities proposed to be taken up for the implementation of MIS/PBS in their respective institutes were prepared and presented by the participants. Many salient points have emerged from the deliberation in the workshop. These include:

1. Conceptual Issues :

- a. The concept MIS should be well understood by all those concerned with its development and use. The basic rationale for an MIS is to improve the decision-making capability of the research managers to manage their resources like people, finance and physical facilities efficiently towards better project management. It is an effective management tool for the research managers which can generate relevant information in a timely way to them for decision-making purposes
- b. As the research institute grows in size and scope of activities, it would be difficult for the research managers to maintain a good grasp of resource use and research programme implementation. An organized MIS can help them to overcome this problem by generating relevant information on the research resources and the manner in which they are currently used in a format suitable for managerial decision-making.
- c. Although MIS is capable of providing information on four categories, viz. projects, people, finance and physical facilities to the research managers to perform the critical management functions of planning, programming, monitoring and evaluation, it may be desirable to focus attention in the beginning only on planning and programming. Once the system is established and the required expertise developed, the inclusion of monitoring and evaluation aspects can be considered.
- d. If appropriate inventories of physical facilities are made available through MIS, replacement and maintenance schedules can be prepared and budget provided. At the present stage of the development of MIS methodology, this aspect is not given adequate attention. However, it is important to include it in future to develop a comprehensive MIS.

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Dr. Jagdeesh C. Kalla, Joint Director and Coordinator, welcoming the participants of the Workshop on Management Information System. Dr Raman and Dr T. Balaguru are also seen.

- e. The basic philosophy underlying the Project Budgeting System (PBS), which is a method of presenting the budget relating to research projects, should be well understood by the research managers and scientists. The PBS attempts to link information on costs, personnel and physical facilities with each research project using a computer-based method. In contrast to financial accounting which is an exact process, PBS involves some form of subjective judgments as to how to allocate common costs to individual projects. It is not meant for accomplishing financial control like the conventional accounting system. It only facilitates research managers to apportion the institute budget among the research projects.
- f. The primary role of MIS/PBS should not be seen as one of policing and control. It only provides the basis for good planning, programming, monitoring and evaluation of research. It strengthens the case for more resources, and also presents the case for effecting reforms at the policy level.
- g. The MIS/PBS should be so developed that it would be of value to various classes of users like policy makers, national level research managers, institute directors, divisional heads, programme leaders and individual scientists.
- h. Uniformity in the basic approach across the research institutes for developing the MIS/PBS should be ensured so that information from them can readily be consolidated at the national level.

2. Technical Issues :

i) *Scientist Time Allocation :*

- a. Since the allocation of scientists' time to various activities like research, teaching, administration/management, and extension is crucial for MIS/PBS development, every effort should be made to ensure the correctness and consistency of the information provided by the scientists. The total time spent by individual scientists on all these activities should add up to 100%. The total time spent on various research projects by the individual scientist should be equal to the time indicated for research.
- b. Furnishing information on time allocation to various activities by the scientists will be new and unfamiliar to them in a majority of the institutes. It is, therefore, necessary for the personnel engaged in MIS/PBS development to interact with individual scientists and explain to them the procedure clearly.

ii) Databases :

- a.** A small amount of information collected for each scientist and each research project permit a wide range of analysis. The MIS/PBS essentially focuses on two types of databases, viz. project and personnel databases. To build these two databases, the required information on research personnel and projects can be collected by using the standard proforma developed by NAARM. Depending on the needs and specific circumstances of the individual institutes, extra fields may be created to accommodate optional information along with essential information in these two databases.
- b.** The personnel database contains two types of information, viz. biographic information useful for manpower planning and career development, and information on how scientists utilize their time which is required for working out project costs.
- c.** The project database should deal with two different aspects of research, viz. scientific aspect and management aspect. Standard lists and procedures should be used to enter information on the name of the discipline, the commodity group, the crop/ species etc.

iii) Project Costing :

- a.** At the present stage of the development of PBS, it will be enough to focus on recurring expenditure that is important for management purposes. Information on non-recurring expenditure or capital costs can be included at a later stage in the development of the system.
- b.** In order to be more realistic in project analysis from a management and economics point of view, it will be necessary to consider the resource personnel costs (scientists, technicians and labour) along with non-personnel costs (materials and supplies, transport and travel, repairs and maintenance, technology transfer, management and administration, utilities and facilities and miscellaneous).
- c.** All these eight cost centres are provided with primary codes. For the PBS in the ICAR institutes, analysis of costs at the primary code level should suffice. If indepth examination of any particular aspect of the budget is warranted by the research managers, it should be possible to disaggregate the budget into secondary and tertiary cost centres. The tertiary codes can also be used to break down the core budget into plan and non-plan, and the external funding into source institution or agencies.
- d.** Whenever the costs on these items related to individual projects are available, it will be possible to allocate the costs to projects on a direct basis. If they are not available, which is the case in most institutes, the budget has to be allocated for

all the items except the scientists to projects indirectly by using a formula. On the basis of the total scientists' time spent on individual projects in comparison with scientists' time spent on total research in the institutes, the costs on these items are proportionately allocated to individual projects. In the absence of sufficient information for direct allocation of costs to individual projects, the use of such formula is the only meaningful option available for PBS development.

- e. In the case of smaller institutes with few scientists, it is possible to consider the actual salary of individual scientists to work out the costs on research, administration/management, teaching and extension on the basis of scientists' time allocation. In larger institutes involving many scientists, it should be enough to use the average salary of a particular grade of scientists to work out these costs.
- f. While arriving at the cost of inservice training of scientists, only the amount incurred on training within the country should be considered. The overseas training cost as well as the cost incurred on consultants from outside should be considered as human capital cost due to the huge amount that is involved. Otherwise, the cost estimation of the project in which that particular scientist is involved will become unrealistic.
- g. Some researchers such as statisticians may be working on a number of research projects of other researchers without having research projects of their own. In such cases, their cost can be allocated to all the projects involved in proportion to the time spent. If kept as a statistics support project, then the cost of such project will be inflated many times particularly when the statistician is in senior position. Depending upon the specific situation, the individual institutes should work out a suitable strategy.

iv) Use of Keywords :

- a. The FAO's AGRIS/CARIS categorization scheme including the keywords descriptor system enable the research managers to identify research projects related to a specific topic in the institute, and make decisions as to whether the overall programme meets the institute's mandate and national priorities. The exhaustive list of CARIS me codes and keywords compiled by NAARM should be able to meet the needs of most of the institutes. However, institute - specific keywords under a given CARIS code can be evolved and used. Such keywords should not duplicate the existing descriptions and should be consistent in their use nationally and internationally. To facilitate this process, the institutes should consult the National AGRIS/CARIS Coordinator who is stationed in New Delhi.
- b. While using the keywords care should be taken not to enter the names of crops or commodities or species or pests and diseases as keywords. In the project database all these are entered as separate fields. Since the keyword may identify only one component of a research project, it should be used with caution in relating it to budget.

2. Management Issues :

i) Commitment :

- a. Senior research managers should be fully committed to the concept of MIS/PBS and should see it as a management tool to increase the efficiency and effectiveness of research in their institutes. They should develop an understanding on and appreciation for MIS/PBS as an effective management tool. Since the development of MIS/PBS is not a one time activity, they should ensure that it receives continued support in terms of providing funds, personnel and facilities for initial development and annual updating.
- b. In order to make the MIS/PBS a viable proposition in the institutes, concerted efforts should be made by those responsible for its development to appraise the senior research managers about the potentialities of MIS/PBS.

ii) Location :

- a. Since MIS/PBS represents a continuing service to management, it requires an institutional 'home' which may vary from institute to institute depending upon the structure. It can be conveniently located as an integral part in the Director's Technical Cell or the Computer Unit. Under the present circumstances, it may not be necessary to establish a separate MIS unit in the research institutes.

iii) Staffing :

- a. The staffing demands for MIS/PBS are modest. Keeping in view the work load and the time required for initial development and annual updating, this work represents only a part-time job at the institute level. Without additional staff being newly recruited, it will be possible for the institutes to develop and maintain the MIS/PBS by redeploying the existing staff.
- b. The strength of the developed MIS/PBS and its effectiveness as a management tool largely depend on the quality of the personnel managing the system. Whatever may be the location, persons with MIS/PBS know-how should be available within the institute. Wherever such expertise is not available, efforts should be made to train the existing personnel to equip them with necessary knowledge and skills to successfully implement the system. Appropriate training for this work need not exceed two weeks. Each institute should have at least one person who is trained to handle MIS/PBS. The institutes should take full advantage of the specialised programmes organized by NAARM on MIS.

(iv) Data Content :

- a. The institutes should neither get 'lost' in a mass of data just because it is available nor become too restrictive and ignore information that may be important for management purposes. Most of the needed information should be already available in the institute itself. The focus should be on essential information for management and decision-making purposes so that the system can be kept small and practical. The system should not be overburdened with information that might be interesting but not necessarily be essential. Initially it is desirable to limit the amount of information included in MIS/PBS until the system operates satisfactorily. As expertise builds up, more information can be incorporated and more detailed analysis can be carried out.

v) Data Collection :

- a. The institutes should collect only the information that are essential for a specific management need and not because they look interesting. Most of the information needed to develop MIS/PBS should be collected from the already existing sources like the payroll, personnel file, budget, project document, etc. Only a minimum amount of new information like the scientists' time allocation on various activities, involvement in research projects etc. should be sought and collected. This will increase the response rate for the scientist questionnaire and enhance the quality of information provided by the scientists.
- b. In order to induce seriousness and ensure collection of complete and correct information from the institute scientists through questionnaire, the top management should provide the necessary moral and physical support for the personnel collecting such information.

vi) Data Quality Issues :

- a. Whenever the information contained in the institute documents like the payroll, budget, personnel file, research project details etc, are not up-to-date and complete, and there are inconsistencies between the different sources, efforts should be made to sort out inconsistencies and information gaps by consulting with the research managers, accounts personnel and individual scientists. The top management should provide the necessary support for this activity.

vii) Equipment :

- a. The recent revolution in information technology was made possible because of the development of Personal Computers (PCs). The development of MIS/PBS requires an IBM compatible PC with a hard disk. Since these relatively inexpensive PCs are already available in majority of the research institutes, it should be possible to develop and use MIS/PBS without involving much expenditure. It also involves only part-time use of the Computer.

- b. It is not necessary to use one particular software to develop MIS/PBS. Any database package can be used. Because of its simplicity, ease of learning and the ease with which a variety of printed tables and graphs can be generated for presentation to senior research managers, the institutes may consider the use of REFLEX software. Although it is a flat database package requiring double entry of certain information in both the project and personnel databases, its files can be translated to and from QUATRO, LOTUS, dBASE, PARADOX and ASCII TEXT. To begin with, the institutes may develop the MIS/PBS using the simple and easy to use REFLEX package which can be converted to relational database programmes later. This is an issue on which each institute would like to make its own decision.

viii) Costs :

- a. Implementation of MIS/PBS will certainly involve costs in terms of staff time, hardware, software and stationeries. Since the entire process involves part-time use of the staff and computer hardware, the benefits will far outweigh the costs incurred.

Workshop on Implications of Farming Systems Research/Extension (FSR/E) in the Indian Context
(Coordinators: T. Balaguru & R. Chittiraichelvan)

Inadequacy of the traditional, disciplinary-oriented research strategies to solve the complex problems of small resource-poor farmers working in less favourable natural environment has led to the evolution of a more holistic and systemic approach, popularly known as Farming Systems Research/Extension (FSR/E). This strategy emphasizes that the research agenda should be determined by explicitly defined farmer's needs through an understanding of the existing farming systems, rather than its perception by the researchers.

Realizing the need to tailor research to meet client demand, many programmes have been initiated in the recent years with an orientation and focus on farming systems. The integration of certain desirable features of the complex FSR/E approach into the mainstream agricultural research; demands initiative and innovativeness on the part of the researchers and the system. In order to provide an opportunity for the senior level functionaries in the research system to develop specific implementable proposals, and to consider ways and means of integrating FSR/E proposals within the existing system and framework of research and extension, a 4-day workshop on 'Implications of Farming Systems Research/Extension (FSR/E)' was organized at NAARM, with the support provided by the Ford Foundation, from 25th to 28th November, 1991.

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Dr J.S. Kanwar, Deputy Director General (Emeritus), ICRISAT, speaking at the inaugural session of the workshop on Implications of Farming Systems Research / Extension in the Indian Context. Dr T. Balaguru, Coordinator of the workshop is also seen.

The specific objectives of the workshop were:

- i) to clarify conceptual and methodological issues of FSR/E;
- ii) to identify modifications in the existing concepts, approaches, organisational structures and operating procedures, which will encourage greater interaction among the researchers and farming community;
- iii) to identify changes in the information feedback system so that insights from the field level could be used more effectively to reformulate research policies and agenda; and
- iv) to develop an implementable plan for integrating FSR/E concept into the mainstream research process.

About 54 senior level functionaries from the Agricultural Universities, ICAR and its Institutes, Non-Government organisations, the Ford Foundation and NAARM participated in the Workshop. In addition, few selected resource persons with a wide experience of working on FSR/E concept in their own countries and elsewhere also participated in the workshop and shared their experiences. The workshop was organized under the following four major themes.

- i) Farming Systems Research/Extension (FSR/E) approaches in Indian Context;
- ii) Implementation of FSR/E approach in India;
- iii) Integrating FSR/E into the existing research system; and
- iv) Successful implementation of FSR/E in India - reliance on appropriate HRD efforts.

Theme papers providing the background materials and highlighting the issues that need to be addressed to develop implementable proposals were prepared by a panel of experts, who have wide experience and insights into the problems. These theme papers were presented by invited speakers which was followed by detailed discussions by the participants in groups. The papers presented by the invited resource persons from other countries also served as the background material and facilitated the group discussions. Several implementable proposals were presented by the groups in the plenary sessions. Three Vice-chancellors who participated in the discussions presented action plans developed by them.

The workshop in its deliberations for four days discussed the four themes and issues placed before it in plenary and group meetings. It also had the pleasure of listening to and discussion on a number of invited papers, which had direct bearing on farming systems research and extension. The recommendations which emerged at the end of the proceedings are, as under.

1. Addressing the production constraints of the peasant farming adopting an interdisciplinary and iterative scientific approach, in a holistic manner comprising all the farmstead activities, and in active collaboration with the farmer is to be recognized as the basic tenet of farming systems research and extension (FSR/E). On-farm and on-station activities of the researcher are complementary, more emphasis being on the farmer participatory on-farm activities. A common vocabulary of the various functions of FSR/E would enable an unified approach as well as help in institutionalizing the process.
2. The target area should be identified on the basis of common environmental parameters, production patterns and management practices for arriving at relatively homogeneous recommendation domains suited to a majority of the potential beneficiaries. Status reports prepared for zonal research activities of State Agricultural Universities, periodically updated for socioeconomic parameters, could help in identifying 'niches' where technology generation would be most rewarding. Information on institutional factors like marketing, transport, credit, pricing, etc. may be put on magnetic tapes. Risk-prone environments and newly emerging irrigation command areas are best suited for FSR/E implementation in the initial stages.
3. Farmer participatory research rather than technology verification on farmers fields should be the motto of FSR/E. Indigenous technical knowledge (ITK) could be a good base for mounting research, which is environment friendly as the local production resources rather than high energy inputs are to be used for productivity upgradation. Socioeconomic analytical tools should be employed for interpreting the results so obtained. Feedback as an instrument of modification of research agenda should be simultaneously recognized. Identification of research needs should invariably have the participation of women farm workers who shoulder almost 50 per cent of the workload in most situations.
4. Considering the financial and manpower implications, an independent cadre of FSR/E personnel was not deemed as essential. A team of researchers and social scientists from existing establishments in different organizations having propensity for rural work may be assigned this task. Policy decisions are essential for this reorientation in working out modalities. It was also felt that the Directorate of Cropping Systems Research of the ICAR should act as a 'nodal' agency responsible for financial support, formulating detailed guidelines for on-farm research and monitoring functions in a 'networking' mode of operation.

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- 5 Career advancement criteria for FSR/E practitioners should be different from those normally applied. An innovative system of rewards and incentives would be required to enthuse them. At the same time, laggard performance would invite corrective measures including punitive action.
- 6 For developing a cadre of researchers with a farming systems perspective, the undergraduate and postgraduate course curricula need be given an orientation towards FSR/E requirements. The existing personnel in the national agricultural research system should be subjected to specialized training in national or international institutions wedded to this philosophy. Even policy makers and administrators need sensitization. Institutions like NAARM, MANAGE, IIMA, etc. may be given this responsibility.
- 7 Mini-workshops may be organized at national level as a follow-up action for deliberating on and evolving detailed guidelines for the various facets of FSR/E working.

Workshop on Monitoring and Evaluation of Agricultural Research Project:
(Coordinator: S.N. Saha)

Based on the feedback available from the Course on Agricultural Research Project Management, the Workshop on Monitoring and Evaluation of Agricultural Research Projects was developed and conducted during October 22-24, 1991.

The major objectives of the workshop were:

- i) to understand the requirement of monitoring and evaluation system in Agricultural Research Projects in ICAR and SAUs;
- ii) to identify the relevant issues and areas amenable to monitoring and evaluating and the information requirements of the same; and
- iii) to work out the methodology for information generation, collection and analysis that will facilitate this process

The workshop was a task oriented activity to evolve the comprehensive methodology for monitoring and evaluation of agricultural research projects. In addition to the exercises and group discussions the following topics were dealt in the workshop viz. Project cycle concept, Network analysis, Benefit cost analysis and Monitoring and evaluation.

The Senior / Principal Scientists of ICAR Institutes and State Agricultural Universities, who had undergone earlier the Course on Agricultural Research Project Management conducted by NAARM, were eligible to participate in this workshop. Preference was given to those who were actively engaged in monitoring and evaluation of Agricultural Research Projects.

The workshop programme had lectures, group discussions, practice sessions and exercises leading to the development of system and methodology for monitoring and evaluation of agricultural research projects. The programme provided excellent opportunities for mutual interaction and experience sharing among the participants and faculty.

Twenty three participants from ICAR Institutes and Agricultural Universities participated in this workshop.

Recommendations

The workshop recommended that in order to implement the research projects effectively the following aspects, viz. Project monitoring; Project evaluation; Organisational structure; and Linkages between monitoring and evaluation and organisational structure have to be taken into consideration for monitoring and evaluation. Monitoring and evaluation issues have been identified and their importance in project cycle has been stressed in this workshop.

Workshop on Productivity and Impact of Research; Scientometric / Bibilometric Measures and Influencing Factors (Coordinator : R.B. Gaddagimath)

During the last three decades, many efforts have been made to promote the use of science and technology and to analyze their impact. In the assessment of productivity of the group and the impact of research, three major elements are of obvious importance the 'inputs' to the research; research 'outputs'; and the 'outcomes' of the research. Outcomes of research are really its impact. For many types of research, the impact is a long-term affair and it is extremely difficult to evaluate the work of a particular research group by such means. The extent of publication of research results is the most obvious and immediate impact of a particular research activity. In attempting to assess the impact of research, the identification of quantitative/qualitative criteria is possible. This workshop emphasized the methods and problems associated in this area with the following broad objectives:

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- i. to expose the participants to measurement tools for evaluating research performance;
- ii. to develop awareness on the use of bibliometric and scientometric applications;
- iii. to recognize the importance of citation analysis in measuring performance of science establishments; and
- iv. to indicate use of bibliometric techniques to study aspects of communication in science.

The programme consisted of a blend of special presentations by the facilitator of the programme Dr. F.W. Lancaster. The following topics were covered:

1. Inputs, outputs and outcomes in the research process; Bibliometric and scientometric approaches.
2. Bibliographic databases, SCI and other tools
3. Productivity: author productivity; institutional productivity; literature growth; Ortega hypothesis
4. Macro and Microbibliometrics: national analysis; institutional productivity and impact; bibliometric vis-a-vis other methods.
5. Diffusion of ideas; informal communication; gate keepers; invisible colleges
6. Interconnections among research groups and disciplines.
7. Obsolescence and scatter of the literature of science.
8. Citation behaviour; bibliographic coupling; co-citation and tracing and application of agricultural science.
9. Bibliometric and scientometric indicators of progress in science and application of agricultural science research indicators for development.

The participants were given exposure to the new areas of assessing the research impact through bibliometric and scientometric methods. An opportunity was given to the participants to evolve a consensus about the indicators and parameters for evaluation. The communication process through different publications which can be used as one of the parameters in evaluating a research group has been clearly drawn, and it was suggested that further work through survey for evolving parameters has to be developed for Indian conditions by the Academy.

Thirty three participants attended the workshop. This comprised 10 faculty members and scientists of SAUs, and ICAR, and 23 library and technical personnel of SAUS and ICAR.

Workshop on Meaningful Learning as a Communication Process (Coordinator : R.K. Samanta)

The Workshop on Meaningful Learning as a Communication Process was intended to stimulate thinking about teaching and learning from a communication perception. It addressed the issue of creating appropriate environment / climate and of organizing content resource for better interaction in the classroom ultimately making thinking possible to achieve excellence in teaching and learning.

The workshop was organized during July 9 - 12, 1991. It was designed to develop conceptual and theoretical understanding about instruction, perception formation, learning and varieties of instructional media used in teaching / training by instructional designers, developers and practitioners.

The overall emphasis was laid on the participatory and experiential learning by the participants and coordinators on the selected topics during the course of the workshop. The broad topics / issues covered during workshop were: perception, communication and learning; visual literacy; instructional media, management, media production and use; organisational renewal and individual change process; positive thinking; and developing communication skills.

Sixteen teaching / training faculty in the rank of Senior Scientists, Professors and Associate Professors representing 10 different ICAR Institutes and four State Agricultural Universities participated in the workshop.

Workshop on Effective Instructional Methods (Coordinator : A. Gopalam)

The workshop was organized from October 29 to November 2, 1992. 15 participants representing the disciplines of Plant breeding, Horticulture, Business management, Agronomy, Entomology, Plant pathology, Veterinary pathology and Social sciences participated in the workshop.

The lecture sessions included Instructional systems and methods, Instructional problem analysis, Integrating instructional aids for classroom instruction, Video based instruction, Instructional aids for teaching effectiveness, Computer assisted instruction, Graphic instructional aids, Programmed instruction, and Principles of effective learning teaching process. A number of practical exercises were included in the workshop.

The participants prepared instructional lesson plans in their areas of specialisation and presented them on the final day. They are produced three video-based instructional modules, namely, Hybrid seed production in sorghum. Uses of overhead projector for effective instruction, and Semester system of education - advantages and disadvantages.

Workshop on Motivation Management and Community Participation in Research and Development (Coordinator : A.U. Gadewar)

The concern for excellence is central to motivation. Personal and environmental conditions encourage creativity and productivity in research and development. But poverty and environmental issues make rural development more complex. Participation of all concerned is therefore, needed to make the issues simpler and easy to solve. Democratic philosophy implies that the members of a community should have a voice in identifying problems, and opportunities for selecting appropriate methods of attaining their desired goals. Community participation, hence, has become a defining characteristic of any developmental programme. Of late, participation is treated as a management technique for use in a wide variety of technical and environmental decisions. Another dimension which has been well recognized in India and elsewhere in the world is to seek community participation through non-governmental organisations (NGOs). This ties between research, extension, development activities and peoples' participation form a continuum of activities which need to be strengthened. In order to strengthen and equip the scientists / research managers and other professional staff working in the ICAR Institutes, SAUs and NGOs in R & B, this workshop was organized with the following objectives during January 20 - 23, 1992.

- * to discuss issues and problems related to motivation management in research and development;
- * to share experiences in using the proven techniques of motivation for successfully managing developmental programme; and
- * to identify various mechanisms of enlisting cooperation and collaboration with non-governmental organisations

Thirty one participants attended the workshop. Dr. Shankariah Chamla, Senior Faculty Members, University of Queensland, Brisbane, Australia, acted as the facilitator of the workshop.

The following topics were covered in detail through various lectures, simulation and discussion methods; Motivation of community / other agencies, Community participation for development action; government and non-government organization linkages and how can we work together for rural development.

The workshop recommended that the participative action management model and group action techniques should be followed in R & D efforts. Various practical methods for motivating self, group and community should be employed to achieve excellence in R & D. Various techniques such as Rapid Rural Appraisal, Participatory Rural Appraisal. Stakeholder analysis, SWOT analysis towards client participation are found to be useful. Leadership styles, role and goal clarity increase the motivation, efficiency of the individual, group and community. More and more exercises - video, structured, semi-structured and simulated exercises help increase participation of the client, and understanding and comprehension of the issues involved.

**National Seminar on Policy Issues in Generation, Reliability and
Retrieval of Database on Indian Agricultural Sector for Social
Sciences Research
(Coordinator : K. P. C. Rao)**

Research in social science is seriously constrained by the non-availability and inaccessibility of relevant data. Even when data are available, their scope and reliability are, sometimes questionable. The statistical base with respect to crop, animal and fishery production is subject to assumptions and approximations.

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Dr Shankariah Chamala, senior faculty member, University of Queensland, speaking at the inaugural session of the Workshop on Motivation Management and Community Participation in Research and Development. **Dr Jagdeesh C. Kalla**, Director Incharge and **Dr A.U. Gadewar**, Coordinator of the workshop are also seen.

Over the last four decades, the methodologies of collection of data in the agricultural sector have undergone several refinements, but they are still considered inadequate to meet the needs of sophistication in modern day research and analysis. Further, only a part of the collected data is published in periodicals and statistical abstracts. Since this information database is limited, often researchers need to use unpublished information. Access to unpublished information is very much restricted. A need to take stock of the availability, retrievability and reliability of different types of data pertaining to crop, livestock, fisheries and forestry sub sectors of the agricultural sector has been felt in view of the newer thrusts and directions in which agro-economic research, as a base for policy planning and development, is moving. The Seminar was conducted from 13th to 15th November, 1991 with the following objectives:

- * to bring the representatives of data generating organisations tete - a -tee with the social science researchers dealing with agricultural sector;
- * to discuss improvements in the methodologies for data collection on agricultural sector;
- * to consider the policy issues concerning the accessibility of unpublished information for research purposes;
- * to identify the gaps in data collection efforts and to take up steps to fill them; and
- * to generate interest in developing data series on National Agricultural Research System, resource allocation patterns and research evaluation parameters.

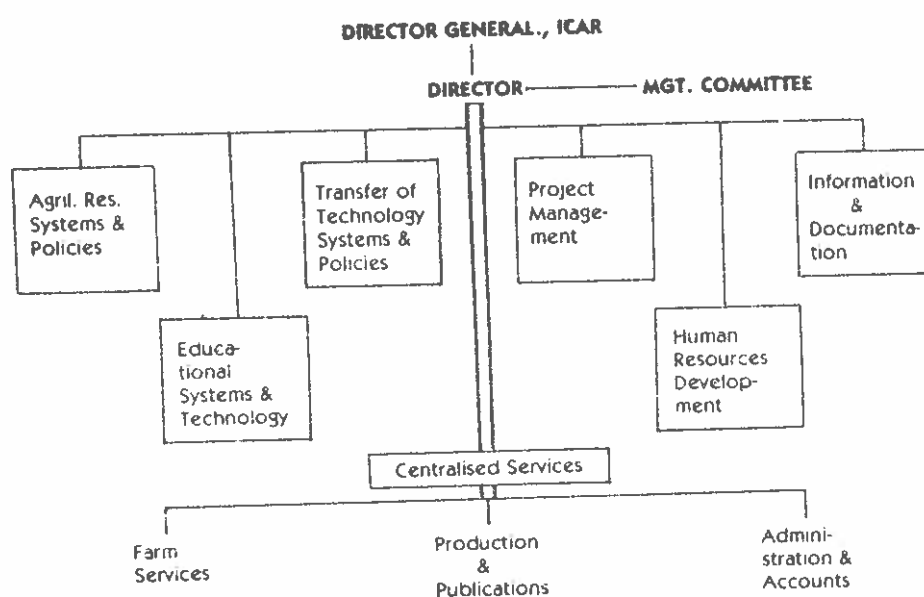
The contents covered in the Seminar include status of database with respect to crops, animals, fisheries, forestry and agricultural research. Among the 27 participants six were from Agricultural Universities, six from data generating organisations, 12 from ICAR Institutes and three were invited speakers. The Seminar has come out with several recommendations regarding collection methodology, strengthening of data generating systems and accessibility of unpublished data to researchers.

Organisational Structure and Salient Activities

The Academy is organised into six subject-matter oriented functional units with a view to impart effective training and conducting research studies in a meaningful way. These are :

1. Agricultural Research Systems and Policies
2. Educational Systems and Technology
3. Transfer of Technology Systems and Policies
4. Project Management
5. Human Resources Development
6. Information and Documentation

All the above mentioned units function with a strong support from the Centralised Services comprising Farm Services and Agro-Technology Blocks; Production and Publications; Administration and Accounts.



AGRICULTURAL RESEARCH SYSTEMS AND POLICIES UNIT

Development of Agricultural Sector is conditioned by resilience and capacities of agricultural research system. For the functions relating to technology generation, gathering of information and procedures involved in agrarian research system would be used jointly for agricultural education, training, extension and client support to bring about desired changes in development strategies. The Agricultural Research Systems and Policies Unit studies the different systems of agricultural research in order to strengthen the various research systems that are essential for agricultural development.

Objectives

- * to study issues on prioritization of research in Agricultural Research Systems and to assist in development of appropriate science policies,
- * to build agricultural resource and research databases for developing suitable research plans and efficient management of National Agricultural Research Systems,
- * to develop systems approach integrating indigenous knowledge with relevant technologies for varied farming situations and constraints,
- * to study National as well as International Agricultural Research Systems and suggest suitable policy alternatives for improvement,
- * to undertake studies on the socioeconomic impact of resource allocation and investment in agricultural research, and
- * to undertake studies on futurology in agriculture and technological forecasting and assessment.

Activities

Research Projects:

1. Database on Crops, Forestry, Animal and Fishery Resources in National Agricultural Research System.

(K. Vidyasagar Rao and P. Kumar)

Data on livestock and fisheries resources collected so far include the following parameters

- i. Population of livestock & poultry during the years 1950-51, 1966, 1971-72, 1976-77, 1979-80, 1985-86.
- ii. Livestock production trends (milk, meat, eggs & wool) between 1950-51 and 1958.
- iii. Data on animal power (1951-57)
- iv. Expenditure on Animal Husbandry & Dairying in different plan periods (1st plan to 7th plan)
- v. India's share in world's exports of various commodities
- vi. Export growth of Indian marine products
- vii. Shrimp exports from India
- viii. Potential of selected fisheries of India
- ix. Wholesale price indices
- x. Global import of leather and leather products (1972-1985)
- xi. India's export performance (1972-73 to 1986-87)
- xii. Availability of hides and skins according to 1986 CLRI Survey.
- xiii. Off-take rates in selected countries (1984)

2. Developing A Suitable Management Information System (MIS) for the National Agricultural Research System' (T. Balaguru and K. Vidyasagar Rao)

Towards developing a suitable methodology in the form of guidelines for the Management Information System (MIS) / Project Budgeting System (PBS), the present study was undertaken with the following objectives:

- i) to develop a suitable MIS to serve the research managers in the research system as a useful tool for managing their research activities, human resources, funds and other physical facilities more efficiently and effectively;
- ii) to familiarize the decision makers in the agricultural research system with the potentialities and use of a well-organized MIS; and
- iii) to generate efficient inferential system from the database.

Appropriate questionnaires were developed to collect information on research projects, personnel and finance. Using this questionnaire, data from one of the ICAR Institutes, i.e., National Research Centre for Sorghum in Hyderabad were collected, analyzed and an organized MIS was developed. The entire material was presented in the form of a case study. Detailed guidelines were also developed on MIS/PBS. These two were used as resource material in the two workshops subsequently organized at the Academy on MIS for Agricultural Research.

Simultaneously, efforts were made to collect data from another institute, National Research Centre on Camel at Bikaner. The collected data is under analysis.

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3. 'Assessment of Returns to Expenditure on Research and Development of Production Technologies for Rice-Fallow Pulses and Chillies'.
(K.P.C. Rao)

Collected data from the Directorate of Economics and Statistics, Government of Andhra Pradesh on Crop cutting experiments of rabi blackgram and chillies for the crop year 1990-91.

4. 'Estimation of Evaluation of Students' Performance at Different Levels'.
(K. Vidyasagar Rao)

During the course of career development in education, it is a preconceived opinion that the previous knowledge gained have direct or indirect impact on subsequent performances. The nature and extent of the influence, however, may vary from individual to individual. In order to verify the above contention, a study was undertaken at graduation level. The data were collected from Andhra Pradesh Agricultural University (APAU), Rajendranagar of 81-85 batch on year-wise and final OGPA. The following results were obtained from the study.

Basic Data

Complete information on all the characteristics were obtained for 50 students only. The analysis of their Socioeconomic background indicates that 2/3 of the students belongs to non-farming and only 1/3 from farming community. Regarding Caste, nearly 2/3 are of other Caste and 1/3 belongs to SC and ST community. Interestingly it was noted that parental occupation and the Caste played no role on the performance at entrance level qualification (intermediate)

Effect of Gender

The total sample is grouped on the basis of gender, i.e., male and female to know the behaviour in relationships among the performance in each sex. The negative and non-significant correlation between inter performance and OGPA for male, female and total indicates independence of these two performance. In other words, intermediate performance is not a complete yardstick to predict the final performance at graduate level. This is an important observation for the administration in education for evolving a criterion for selection of the candidates. During graduation, the performance at earlier periods (1st year) are less correlated as compared to later periods (3rd and 4th years) with the overall performance (OGPA) in case of male students. But in females the performance at 1st year has negative correlation with OGPA, but the relationships improved linearly over subsequent period. This indicates females are highly vulnerable over their counter sex to sudden change in curriculum in the early period.

Effect of Standard of Learning

In order to know the effect of standard of learning on the degree of relationship among the performances, based on performance in entrance level examination (intermediate), the students were classified into top 27% as high or top learners, bottom 27% as bottom or low learners and remaining as middle or average learners. The correlation between inter and during graduation performance in case of top learners as compared to low learners. This means that performance at intermediate has low predictive value in case of top learners as compared to low learners. This is an important note to the administration in the field of education. The reasons for this needs further systematic study. From the average performance levels among the above three groups at inter and graduation levels, it was clear that the linear relation between average performance and standard of learning for intermediate, reflected true evaluation and marking as per the standard. But in B.Sc (Ag) a slight deviation from linearity in particular between middle and top learners leads to a point of doubt on the allotment of grades objectively.

Variability in Performance

The performance in education of an individual can be visualized as an additive effect of two components, as:

$$\text{Performance} = \text{Hereditary} + \text{Environment}$$

In the field of education, hereditary component includes inherent intelligence (IQ) and the environment component consists of curriculum set-up, standard of Institute, where the student was studying, type of examination and evaluation system, parents involvement, etc. It is inferred that the variability among the female students was less over males at the entrance level examination. The variation within males and females increased as the course advanced, but the variability maintained at constant level during later periods (3rd year onwards). No improvement in reducing this variability within the group in subsequent years of study indicates lack of proper attention by the faculty, or lack of proper interaction between student and teacher, or defect in curriculum. A separate study in detail is needed to find out the real factors responsible. This is an important observation to the University administration for formulating future curriculum.

Conclusions

1. The parental occupation or caste had no effect on the performance at entrance level examination.
2. The correlation between OGPA and intermediate performance were very low among male and female students.

3. The predictive value of intermediate was slightly improved in case of low learner students as compared to top learner group.

4. It was shown that the allotment during graduation was not as per the level of learning standards.

5. The variability in performance within the male and female groups were of the same order throughout the course.

Workshops and Seminars

1. Two Workshops on Management Information System for Agricultural Research were organized, one in June (10-20) and another in September (3-13), 1991.
2. Workshop on Implications of Farming Systems Research / Extension (FSR/E) in the Indian Context was organized from November 25 to 28, 1991.
3. Seminar on Policy Issues in Generation, Reliability and Retrieval of Database on Indian Agricultural Sector for Social Sciences Research was organized from November 13 to 15, 1991.

Courses:

1. Course on Computer Application in Agricultural Research and Education was conducted from July 16 to 27, 1991.
2. Course on Training of Trainers was conducted from October 15 to 26, 1991.

Course Material Developed:

1. Agricultural Research Systems of the World.
2. Database Management Concepts.
3. Brain Storming Material.
4. MIS Guidelines.
5. Adult learning theories; Experiential learning; Role of trainers; Training styles and their effectiveness; Learning climate setting; Training needs assesment; Training methods; Facilitation skills; Managing learning activities; Training programme development and implementation.

EDUCATIONAL SYSTEMS AND TECHNOLOGY UNIT

The primary function of this Unit is to render assistance to the SAUs and ICAR Institutes dealing with education and provide support to develop better teaching techniques. Education Technologists interact closely with Subject Matter Specialists to develop viable and useful courses and teaching techniques.

Objectives

- * to organize teaching and training programmes in higher agricultural education with special reference to curriculum development, instructional systems, production of learning resource materials, evaluation and assessment, and educational administration,
- * to undertake research projects and case studies for evolving more effective and efficient approaches, methods and models in higher agricultural education system,
- * to develop information database on education system in SAUs and ICAR Institutes,
- * to provide technical support to SAUs and ICAR Institutes on educational matters to enhance their teaching-learning effectiveness, and
- * to establish linkages with international bodies imparting higher agricultural education.

Activities

Courses/Workshops:

1. Course on Curriculum Design and Development was organized from September 17 to 28, 1991. Seventeen participants from different Agricultural Universities attended.
2. Orientation Course on Fisheries Extension was organized from November 8 to 23, 1991 for the benefit of the fisheries scientists and fisheries extension personnel working in various states. Sixteen fisheries extensionists attended.
3. Workshop on Effective Instructional Methods was organized from October 29 to November 2, 1991. Fifteen faculty members from State Agricultural Universities participated.

Course Material Developed:

1. ADII model for analysing the instructional problem.
2. Computer aided instruction.
3. Programmed instruction.

TRANSFER OF TECHNOLOGY SYSTEMS AND POLICIES UNIT

Development of appropriate transfer of technology (TOT) systems is important to improve the productivity of researchers, extensionists and farmers. The effectiveness of these systems depends on various factors such as managerial efficiency of the managers, span of control, communication strategies, planning and execution. Further, the informal and institutional linkages are the key determinants of success of the TOT systems. Identifying these linkages is a major mandate of this Unit.

Objectives

- * to study the projects and programmes of technology generation, and evolve tools for their assessment in transfer of technology,
- * to identify the managerial constraints in existing systems, and develop suitable and integrated approaches for effective transfer of technology,
- * to suggest suitable policy framework for effective implementation of technology, and
- * to evolve an efficient agricultural communication system to support the aforesaid activities.

Activities

Research Projects:

1. Impact of NAARM Training in Agricultural Research Project Management on ICAR Scientists and their Training Needs.

(R.K. Samanta, R. Chittiraichelvan and A.U. Gadewar)

In order to improve the design, pattern and content of training programmes conducted at the Academy, this project was undertaken to evaluate their overall impact in terms of instructional outcomes, job performance, benefits to client institutes and training needs. The study was undertaken with the following objectives:

- * to evaluate the impact of NAARM training on ICAR scientists and their environment.
- * to find out how much has the NAARM training helped the scientists perform their jobs better in their working institutions.
- * to find out how much have the client institutes benefited from their scientists trained at NAARM.
- * to document the strengths and weaknesses of the existing training programmes.
- * to evaluate the training needs of the ICAR scientists.

A total of 800 scientists with 500 of them under the category of scientists trained at NAARM in any one of the training programmes during the period 1976-87; another 200 under the category of peers (non-trained) of the trained scientists working in the same institutes; and the rest 100 fall under the category of Heads of Divisions / Directors.

Some of the salient points emerged from the study include:

- * The trained scientists have better appreciation and understanding of the Agricultural Research Project Management (ARPM) training than their counterparts who have not been trained in any of the training programmes. Therefore, all categories of scientists, irrespective of their cadre in ICAR, must undergo ARPM training at the Academy which may help them in understanding and managing agricultural research projects better.
- * ARPM training has been able to induce team spirit, leadership and fraternity among the scientists. This significant aspect can be exploited suitably by the institutes for increasing research productivity of both the scientists and institutes to bring about overall institutional development.
- * The Field Experience Training should receive more importance and emphasis in the design, content and implementation of ARPM training.
- * The existing evaluation system for the Foundation Course may be modified suitably to include continuous and effective monitoring of the trainees over the entire period of the training programmes.

- * Post-training evaluation and follow-up of the scientists should be undertaken at the concerned institute level and by the Academy to get the feedback towards improving the training programmes and better utilization of scientists' skills for research productivity and institutional development.
- * Course curriculum and the contents of various topics require modification to keep pace with the changing needs of the scientists and institutes.
- * Due care may be taken in recruitment and development of NAARM faculty by considering the needs and suitability of the positions, and nature of the job.
- * A conducive climate and infrastructural facilities may be created in all ICAR institutes, so that things learnt in ARPM training at the Academy can be utilized by the scientists in a more meaningful way.
- * In dealing with management topics, like research project formulation and implementation, the network analysis and management process examples may be drawn from Indian experience with agricultural bias with the support of case studies.
- * ARPM training should be need-based and relevant to the working situation of the scientists researching in various projects and locations of ICAR institutes.

2. Evaluation of Use and Impact of Sorghum Publication - Nutritional Disorders of Sorghum.

Based on the recommendation of the International Symposium on Sorghum in the 1980s, Australian Centre for International Agricultural Research, Australia Commissioned a collaborative project between the All India Coordinated Micronutrient Programme of the Indian Council of Agricultural Research, and the Plant Nutrition Group of the Department of Agriculture, of the University of Queensland. In addition, the United States Department of Agriculture and the University of Nebraska and ICRISAT participated in this programme.

As a result of the collaborative project, the book entitled "Nutritional Disorders of Grain Sorghum" was co-published by ACIAR and ICAR and the University of Queensland in 1988. In 1989, the ACIAR commissioned Dr Shankariah Chamala, Senior Faculty Member of the University of Queensland to undertake the evaluation of the book with the following specific objectives:

- * To find out the extent of distribution of the book throughout India more particularly in the sorghum areas.
- * To find out who are the recipients of the publication, such as Sorghum Researchers, Extension Personnel, University Lecturers, libraries (postgraduate and undergraduate users).
- * To find out its impact/usefulness as judged by users in their research, teaching and extension activities.
- * To evaluate the impact of this publication on any production oriented activity.

The University of Queensland involved the National Academy of Agricultural Research Management, Hyderabad in designing and developing the methodology of the project to be carried out in India.

Methodology

Major sorghum growing states of India such as Andhra Pradesh, Karnataka, Maharashtra, Tamil Nadu, Gujarat, Haryana and Madhya Pradesh were chosen.

A three to seven points semantic scale was developed to seek information on various aspects and for the quantification. The questionnaire were sent to scientists, academics and librarians. Out of the 213 questionnaires mailed 73 were responded. The breakup of the questionnaires is Scientists 110 mailed 44 received. Academics 79 and 21; and Librarians 24 mailed while 8 received.

Salient Findings

The responses indicated that the users used publication from "some time" to "frequently". From among the users maximum users belong to the disciplines of soil science, agronomy and plant breeding. They were from research as well as teaching profession. Also, among the users there were postgraduate and undergraduate students.

With regard to usefulness of the publication, about 50% of the scientist-respondents perceived the publication as highly useful. However, the respondents have revealed that they read only relevant portions contained in the book. The Scientists and teaching faculties' perception of the likely usefulness of the publication in a few distinct areas of their responsibilities have a direct

bearing on the utilisation of the book overtime. In addition Scientists stated that this publication has usefulness in preparation of lecture notes (52%), advising extension functionaries (52%), and advising farmers (45%). Similarly, teaching faculties rated this publication as likely to be useful in research (76%), guiding Ph.D Scholars (81%), Extension activities (81%), and training activities (76%).

A little over 60% of the respondents have expressed the possibility that the publication might be of help in developing the interdisciplinary team approach. The respondents recommended that similar type of publication may be brought out in oilseeds, bajra, pulses, maize and horticultural crops.

Workshops:

1. Workshop on Meaningful Learning as a Communication Process was organized from July 9 to 12, 1991 for Professors, Associate Professors, Senior Scientists from SAUs and ICAR Institutes.
2. Workshop on Motivation Management and Community Participation in Research and Development was organized from January 20 to 23, 1992 for faculty members of SAUs, ICAR Institutes and Non-Government Organizations.

PROJECT MANAGEMENT UNIT

This Unit is designed to integrate complex efforts with regard to planning, scheduling, directing and controlling institute's resources for a project, established with specific goals and objectives. It provides an individual research worker the necessary tools and techniques which can help him find proven and workable solutions to problems that may occur in the management of a project.

Objectives

- * to evolve an appropriate methodology for identifying relevant agricultural research projects in a given thrust area and to develop criteria for their appraisal,
- * to develop and formulate a holistic/systems approach for project planning and management
- * to study issues, develop case studies and evolve suitable tools for carrying out the activities related to project cycle more effectively, and
- * to develop a database system on agricultural research projects needed to fulfil the above objectives.

Activities

Research Projects

1. Development of Appropriate Criteria for Project Appraisal in Basic and Applied Research in Agriculture.
(V. Raghavendra Rao)

Work on this project has been initiated and the preparation of proformae is in progress.

2. Monitoring and Performance Evaluation of Agricultural Research Projects - NARP.
(S.N. Saha)

Workshop has been scheduled during November, 1992. Seven centres of NARP and four centres of ICAR Institutes have been chosen for detailed case studies and the work is in progress.

3. Management of Agricultural Research Stations: Cases (K. Venkateswarlu)

The project is undertaken with the following objectives:

- to discuss the important factors for establishing a research station,
- to discuss the planning and development of research stations,
- to exchange the practical experiences of farm operations management through case studies,
- to discuss the issues related to strengthening the agricultural research support system,
- to develop farm operations management models for different types of research stations,
- to discuss the issues related to agricultural research management.

The work is in progress.

Courses/Workshops:

1. Programme on Administrative and Financial Management was organized from September 2 to 6, 1991 for the Senior Executives including Deputy Directors General, Assistant Directors General and Directors of ICAR Institutes.

2. Workshop on Monitoring and Evaluation of Agricultural Research Projects was organised from October 22 to 24, 1991 for the Senior / Principal Scientists of ICAR Institutes and State Agricultural Universities.

Course Material Developed

1. Project Appraisal for Fisheries Extension Education.
2. Project Concept
3. Four cases on PERT and CPM.
4. MIS Guidelines.
5. Case Study, Role Play; Co-training, Training Evaluation, Setting Training Objectives, Lecture Method, Method Demonstratin, Training Programme Logistics.

HUMAN RESOURCES DEVELOPMENT UNIT

The major thrust of this Unit is in the use of experiential learning. Structured experience for training in human behaviour is extensively used for self-improvement by trainees, both in the theory and in the organisational behaviour practicals. The concern in all the training programmes is the understanding of human behaviour with a view to changing, direct and control behaviour. Emphasis is on group behaviour in addition to individual behaviour. There is a constant endeavour to bring about a general appreciation on the importance and usefulness of effective management of human resource, which is the most important resource of any organisation.

Objectives

- * to evolve effective and efficient approaches, methods and models in different areas of human resources development such as interpersonal relationships, motivation, group dynamics, team-building, conflict management, leadership, organisational climate, institution-building and personnel management,
- * to develop information database on various aspects of personnel management in the ICAR Institutes and SAUs for use in manpower planning, recruitment, training, performance appraisal and career development, and
- * to undertake systematic review and study of personnel management problems of ICAR Institutes and SAUs.

Activities

Course:

Two Courses on Human Resources Management were conducted during August 20 -31, 1991 and December 10-21, 1991 for Senior and Principal Scientists of ICAR Institutes and Professors, Head of the Departments and Principals of State Agricultural Universities and Bangladesh nationals.

Resource Material Developed:

1. Transactional Communication and Negotiation for Improving 'Perceptions' and 'Team Building'.
2. Transactional Analysis.
3. Personal Effectiveness as Research Managers.
4. Understanding Human Behaviour.
5. Stress Management.
6. Organisational Effectiveness
7. Organisational Development
8. Theories of Organisation
9. Effective Presentation

INFORMATION AND DOCUMENTATION UNIT

Information and documentation form the most important inputs in any scientific activity. Organized scientific information base is important for decision-making at all levels. Personal scientific information management and retrieval is necessary for every research worker. The Information and Documentation Unit provides faculty support for the training programmes of the Academy in addition to maintaining an excellent Library and Documentation Centre.

Objectives

- * to build a resource Library and Documentation Centre on agricultural research management and educational technology,
- * to set up a clearing house for user education/instruction, research on documentation and scientific information management,
- * to assist in developing a policy framework for modernization of libraries through research on and feedback from the present systems, and
- * to establish an information network of agricultural libraries in India.

Activities

Library:

The Unit has established a well-organized specialized library. The resource library holds documents on Agricultural Research Management, Science Policy, Planning, Futurology, Forecasting, Project Management, Educational Technology, Technology Systems and Policies, Women Studies, Research Communication, Information Management, Computer Technology, etc. International Corner and Foreign Tour Reports Section is a unique feature. The Library now holds about 21,000 documents and subscribes to about 425 journals.

Reprographic Services:

Reprographic section of the Unit is equipped with two Finesse 4515, one Nashua, one UBIX 1800Z and Toshiba BD 704 plain paper copiers and three scanning machines for meeting the photocopying and reproduction facilities. Micrographic section has one Minolta RP 505 and AARQUE microfilming unit.

Computerization:

Computerization of the library holding is in progress, and major collections are accessible through computer. Journals subscription is also in the process of computerization. The Unit has PCAT (40m) and UNESCO's CDS/ISIS package version 2.32 for information storage and retrieval. Three separate databases for CAT (books and serials), ARMA (Agricultural Research Management Abstracts) and FTR (Foreign Tour Report) were developed.

Information Resource Laboratory:

Some of the agricultural databases are being procured on CDROM which would be used in training and in providing services for the scientists. The CDROM work stations are also being established and plans are on for networking the user services. A few CDROM demo disks are available with the Unit for information retrieval. On procuring the CDROM disks (archival and current), the services will be extended to all scientific community.

GISTNIC Information Booth:

The National Informatics Centre (NIC) has installed general information booth, a dial-for-data service. The General Information Service Terminal of National Informatics Centre (GISTNIC) makes available basic information about India, important national and international information, static or non-static information, railway/air timetable, recreational and tourist information. GISTNIC terminal at the Academy is extensively used for supplying statistical information. Efforts are also being made to access bibliographical databases available at NIC and also for E. mailing.

Agricultural Research Management Abstracts (ARMA):

This is a unique halfyearly secondary publication brought out by the Academy broadly covering the areas related to agricultural systems and policies, project management, transfer of technology, educational technology, agricultural economics and information management. About 425 scientific and technical periodicals are scanned and significant papers are indexed and abstracted. The computer database on research management was developed for an effective storage and retrieval for the benefit of research managers, educationists and decision-makers. About 700 copies were distributed for organisations involved in agricultural research, administration, management and policy making. At present, the publication is distributed on exchange basis. Reprographic facilities for the original documents appearing in the abstracts are also offered at nominal cost.

Courses/Workshops:

1. Course on Information Management and Research Communication in Agriculture was organized from November 26 to December 6, 1991 for the benefit of the faculty, scientists and technical personnel.
2. Workshop on Productivity and Impact of Research: Scientometric/Bibliometric Measures and Influencing Factors was organized from June 25 to 29, 1991 under the FAO/UNDP Advanced Centre on Educational Technology and Research Management in Agriculture.

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UNDP Advanced Centre on Agricultural Education and Research Management

An Advanced Centre on Agricultural Education and Research Management was established in 1987 with FAO-UNDP-UNESCO support. The Centre organizes workshops, seminars, training programmes and courses in educational technology and agricultural research project management. The Centre has acquired sophisticated equipments to impart effective training in these areas. Several faculty members have received advanced training in various Institutes and Universities abroad, and a number of overseas consultants visited the Academy to share their expertise with the faculty, under their support. The State Agricultural Universities and ICAR Institutes are effectively utilizing the Centre for strengthening their curriculum and instructional programmes. The Centre can now rightfully extend its activities to the entire South Asia and Southeast Asia region.

Prof. F. Wilfred Lancaster, University of Illinois, Urbana Champaign, USA, was at the Academy from June 10 to July 6, 1991 as a Consultant. Under this Advanced Centre, he acted as the Facilitator for the Workshop on Productivity and Impact of Research : Scientometric / Bibliometric Measures and Influencing Factors.

Dr Shankariah Chamala, a senior faculty member in Agricultural Extension and Management at the University of Queensland, Brisbane, Australia, was at the Academy from January 20 to February 7, 1992 as a Consultant under this Advanced Centre. Dr Chamala had interactions with the faculty members in areas like transfer of technology, human resources management, farming systems research, project management, systems approach, educational video programmes, etc.

Under the UNDP Project, four faculty members completed advanced study in various Institutions and Universities in USA.

CENTRALIZED SERVICES

PRODUCTION AND PUBLICATION SECTION

NAARM has taken up several publications on its own to help the research managers to keep abreast with the latest techniques of management. The Academy itself publishes the materials developed by the faculty for training purposes.

To facilitate the production and publication activities, the Academy is equipped with the Desktop Publishing system from Apple Macintosh, Reprographic section comprising Offset Printing Press (Swift 300 Deluxe), Plate-making and Binding systems. The unit also has screen printing facility.

Some of the publications printed at the Academy include:

1. Course materials and reading materials for each course conducted by the Academy.
2. Case materials prepared on the basis of research studies undertaken by the faculty.
3. Under the new series of publication 'Occasional Papers', started last year, a second paper entitled 'Agro-Forestry Education: A Learner - Centred Approach' was published. The Occasional Papers critically examine current and arising issues in the area of research management, educational technology, administration and management as well as newer approaches to research planning and prioritization.

Besides, the Academy brought out the following periodical publications:

Agricultural Research Management Abstracts (ARMA) a half yearly publication appearing every June and December.

NAARM News, published every quarter, brings out the information about the events/programmes, both present and future, and welfare activities of the Academy.

NAARM Alumni News, a halfyearly publication, brings out information for and about the alumni of the Academy.

FARM SECTION

The campus of the Academy is being developed with an approach to functionality and aesthetics. The landscaping is done with a view to providing a congenial environment for academic activity. Stress is being laid on utilizing a large and varied collection of trees, shrubs, climbers, herbaceous plants and ground covers with an eye on the educational value of a well-labelled rich collection of flora. The Agro-technology Demonstration Blocks have conceived as a show window of the National Agricultural Research System. The latest technologies and crop varieties released by different ICAR Institutes and Agricultural Universities are put on show in one place for the benefit of trainees and visitors comprising national and international scientists and policy makers of agricultural research and development.

Objectives:

Landscaping the campus, developing and maintaining the estate, and laying out the Agro-technology Demonstration Blocks in different areas of crops and different silvi systems are the objectives of the Unit. The landscaping activities cover entire 50 hectares of the campus. The Agro-technology Demonstration Blocks cover an area of 30 hectares out of which eight hectares are covered each by field crops, orchards and vegetable cultivation. The remaining area is covered by silvi systems, namely, agri-horticulture, silvi-pasture, horti-pasture, agro-forestry, agro-energy systems and plantations like high density plantations, graded height plantations, green belts and tree museum.

Activities

Field crops:

Eleven varieties of different field crops like maize, sorghum, sunflower and redgram were grown during the Kharif (1991) and Rabi (1991-92) seasons. These include the currently recommended and latest varieties released by ICAR institutes and agricultural universities. Emphasis was given to maximise the productivity with optimal utilisation of available resources.

Horticultural crops:

Demonstration of fruit crops, vegetable crops and flower crops. The fruit crops included several improved varieties and hybrids of guava, sapota, mango, grape, ber and pomegranate.

Nineteen kinds of vegetables in 35 varieties (transplanted and direct sown) were grown in kharif (1991) and rabi (1991-92) seasons. The Floriculture Block included several varieties of roses and jasmine.

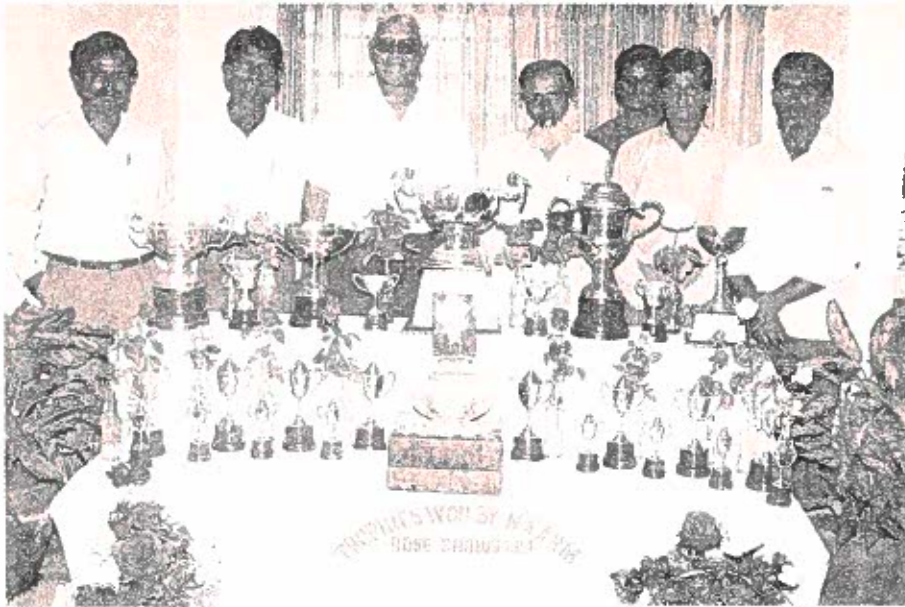
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The Academy had a rich haul of prizes at the rose shows conducted in the twin cities.

91-92

Landscaping:

Importance was given in making the campus green by establishing and maintaining lawns in and around main gathering places like Administrative Block, Academic Block, Faculty Block, Halls of Residence, Scientists Home and Stage area. Several flowering annuals, borders, hedges and foliage shrubs were added to the existing ones provide aesthetic environment to academic activity.

Silvi Systems:

Besides the conventional input systems for the increased productivity, several systems have been established in order to demonstrate the increase in productivity per unit area of land by way of obtaining food, fuel and fodder from the same piece of land. These include agro-horticultural, agro-forestry, horti-pastoral, silvi-pastoral and energy plantation systems which were developed on low moisture and poor fertility status soils. Fruit species like custardapple, phalsa, amla and woodapple were included in these systems.

Graded height green belt plantations:

Several useful forest species were established in peripheral and other areas to provide aesthetic environment value to the visitors and trainees in the campus.

Honours and Awards:

It was a rosy year for the Academy at the two rose shows held in twin cities. An entry at the Secunderabad rose show was judged as King of Show. In the Hyderabad rose show also the Academy did very well and won the coveted Hyderabad Rose Society Challenge Cup. At the two prestigious shows, awards were showered like confetti on the entries from the Academy.

The Academy reigned supreme with an aggregate of 18 points and was awarded the Horticulture Society Trophy at the Secunderabad rose show. The bag of prizes at the Hyderabad rose show included the Hotel Sarovar Challenge Cup, Rosarian Institute of the Year Award, the Srimathi D. Indira Memorial Challenge Silver Trophy for Prince of the Show and Hyderabad Rose Society Challenge Cup besides winning eight first, eight second and two third prizes.

ADMINISTRATION AND ACCOUNTS

The Administrative Wing of the Academy is organized into the following five sections to provide effective service:

1. Admn. I Section --- Establishment matters
2. Admn. II Section --- Cash and Bills
3. Admn. III Section --- Stores and Purchase
4. Admn. IV Section --- Estate Management & Training
5. Audit and Accounts --- All financial matters
Section

The Administrative Wing, besides ensuring smooth conduct of various training programmes of the Academy, also provides faculty support in the areas of Administration and Accounts in the training programmes.

The Human Resources

The category-wise position of human resources during the period is set out in the following Table

Name of the Post/ Category	Total Sanctioned Strength	Posts filled as on 31.03.1992
Scientific	43	21
Technical	41	39
Administrative	49	41
Auxiliary	28	21
Supporting	37	36
	198	158

FINANCIAL OUTLAY

Sl. No.	Head of Account	Rupees in hundreds			Rupees in thousands		
		Actuals during 1990-91			Revised Estimates 1991-92		
		Non-plan	Plan	Total	Non-plan	Plan	Total
1.	Pay & Allowances	57,76,400	12,36,500	65,12,900	67,75,000	35,50,000	1,03,25,000
2.	Travelling Expenses	1,58,200	2,43,700	4,02,900	1,25,000	15,00,000	16,25,000
3.	Recurring Contingencies	30,48,800	6,71,200	37,20,000	20,00,000	8,00,000	28,00,000
4.	Non- Recurring Contingencies						
	A) Equipments	-----	22,78,700	22,78,700	-----	16,50,000	16,50,000
	B) Works	-----	35,69,900	35,69,900	-----	15,00,000	15,00,000
	C) Others	-----					
		84,84,400	80,00,000	1,64,84,400	89,00,000	90,00,000	1,79,00,000

MISCELLANEA

HINDI DAY

Hindi Day was celebrated at the Academy on September 13, 1991. Inaugurating the celebrations Dr V. R. Chakravarthy, Professor and Head, Department of Hindi, Osmania University, said that it is important to learn Hindi as a language for day-to-day use rather than learning it as a literary form; when this kind of approach is adopted there are more chances of it being accepted as a language spoken by all.

Dr K.V. Raman, Director, said that for a nation like India, national language is important for its development and hence, Hindi should be promoted. He promised that more use of Hindi in the Academy's programmes will be taken up.

Dr Raman distributed the prizes to the winners of elocution and essay writing competition. Dr Devesh Kishore, Chairman, Official Language Implementation Committee and Principal Scientist, elucidated the various schemes the Academy has taken up to popularize Hindi.

Mr D.Venkateswarlu, Technical Officer, welcomed the gathering while Mr Pradeep Singh, Hindi Translator, proposed vote of thanks.

PHYSICAL FACILITIES

The Academy has established the following laboratories to make the teaching - learning process more effective, and to help the trainees learn effective use of mass media for dissemination of information.

Video Instruction and Production Laboratory

This laboratory, comprising two television studios and a Master Control Room, facilitates intensive training of scientists in the preparation of effective resource materials for telecasting. It helps in conducting role play exercises for personality development. Video educational programmes are also produced. It is also useful for proper speech communication skills.

Audio-Visual Lab and Photography Unit

The Unit, equipped with all modern photographic and audio reprographic facilities, provides an excellent opportunity to gain experience in the preparation and use of audio-visual aids for effective presentation. These include Overhead Projectors, Epidiascope, Coloured Transparency- making units, Slide projectors, Tape recorders and Public Address Systems.

Speech and Mass Communication Laboratories

These laboratories facilitate the scientists to record their programmes such as interviews, discussions and talks, specially for the rural listeners.

Organisational Behaviour Laboratories

Individual and group exercises in team-building, leadership and group dynamics conducted in these laboratories provide the trainees an opportunity to understand the various aspects of organisational behaviour including human thought, action and behaviour.

Self-aided Instructional Laboratory

This laboratory provides opportunity for the participants to undergo structured and programmed instruction under self-learning situations. Automated electronic equipments and computerised instruction facilitate the process. A set of carrels with Telex Caramate imparting a tape slide programme, a VCP with monitor for a video-tutorial and a computer for a programmed instruction. Software developed for learning Plant Pigments/ Chlorophylls is being evaluated for its effectiveness as a learning tool.

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Information at finger tips! GISTNIC terminal at the Academy.

Computer Cell

A Computer of mini-frame WIPRO S-306 with 4 MB RAM and 170 MB hard drive along with 12 terminals having 640 K RAM was installed. In addition to this, a DCM TURBO 2000 PC-AT forms a part of the cell. Three dot matrix printers and two Near Letter Quality printers besides one plotter are also acquired. These systems are used to provide intensive training on computers to the trainees of FCARPM, prepares Databases and develop Library Information Service. Besides, computers are also being used in Office Administration for preparation of Paybills, Accounting & Budgeting, Recruitment Processing and secretarial work. The Faculty and Staff are also being trained on the use of various application packages such as Wordstar, dBase and Lotus with a view to increase work efficiency.

Desktop Publishing System

A Desktop Publishing System, Apple Macintosh-SE was installed with several software packages including MS Word, Mc Write, Mc Paint, Mc Draw and Hindi Font. These packages would help improve the quality of publications at the Academy.

GISTNIC Services

The General Information Service Terminal National Informatics Centre (GISTNIC), a dial-for-data service provided by the National Informatics Centre, was installed for the benefit of the trainees and the faculty of the academy. This service provides basic information about India, important national and international information, infrastructural and recreational facilities and static or near static information like railway timetable, recreational and tourists information. GISTNIC also offers a few other databases in specialised areas for more specialised users as administrators, economists and financial analysts. The terminal is also used for accessing other online databases and electronic mail.

Publications

Books

1. Raman, K.V. and Balaguru, T. (Ed) 1992. Farming Systems Research in India : Strategies for implementation. Proceedings of the National Workshop held at NAARM from November 25 to 28, 1991.
2. Samanta, R.K. 1991. Manual on Instructional Aids for Teaching Excellence, Mittal Publishing House, New Delhi.
3. Samanta, R.K. 1991. Agricultural Extension in the Changing world perspective, Uppal Publishing House, New Delhi.

Scientific / Review Papers

1. Chittiraichelvan, R. and Raman, K.V. 1991. Indigenous knowledge for farmers : Its use in extension strategies for rainfed agriculture (Ed. Singh, S.P. and Prasad, C.). Indian Society of Extension Education, International Fund for Agricultural Development, Indian Council of Agricultural Research, pp. 185-200.
2. Gopalam, A. and Devesh Kishore, Self-Aided Instructional Laboratory. Proceedings of International Conference on Experiential Conference held at University of Pondicherry, India from March 9 to 11, 1992, pp 149-152.
3. Gopalam, A. and Raman, K.V. 1992. 'Application of an Expert System for the Evaluation of Agricultural Research Projects in Indian Context of Agricultural Research'. Innovation in Management Development published by Tata Mc Graw-Hill Publishing Company Limited :384-393.
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FACULTY DEVELOPMENT

Participation / Presentation of Papers

1. Balaguru, T. participated in ASCI-NAARM-ODI Asia Regional Workshop on 'NGO's, Natural Resources Management and Linkages with the Public Sector, from September 16 to 20, 1991 held at Administrative Staff College of India, Hyderabad and presented a case study on 'Role of NGO's in Krishi Vigyan Kendras (KVKs).'
2. Chittiraichelvan, R. attended an International Conference on 'Extension Strategy for Minimizing Risk in Rainfed Agriculture' organised by the Indian Society of Extension Education and ICAR in collaboration with International Fund for Agricultural Development from April 6 to 9, 1991 at New Delhi.
3. Devesh Kishore, chaired a session at the Workshop on 'Aquaculture Extension Research Needs ' organized at the Central Institute of Fresh Water Aquaculture, Bhubaneswar on June 20, 1991. He presented a paper on 'Principles and Systems of Diffusion of Aquacultural Technologies leading to Adoption'.
4. Devesh Kishore, participated in the International Conference on "Experiential Learning from March 9 to 15, 1992 held at Pondicherry. He also presented a paper on 'Self-aided Instructional Design' and chaired a session on Instructional Systems and Design.
5. Gadewar, A.U. attended the 'Asia Regional Workshop : NGOs, Natural Resources Management and Linkages with the public sector'. held at Administrative Staff College of India, Hyderabad from September 16 to 20, 1991.
6. Jagdeesh C. Kalla. presented a paper on 'Coping strategies of households in high risk, arid environment: a case study of drought, 1987, in Arid districts of Western Rajasthan at IIM, Ahmedabad from March 9 to 12, 1991.
7. Jagdeesh C. Kaila, attended a Workshop on Methods and Procedures for Comparative Advantage Analysis for CGPRT Crops' held at Bogor, Indonesia from November 4 to 9, 1991.

8. Jagdeesh C. Kalla, participated in the International Conference on Experiential Learning from March 13 to 15, 1992 held at Pondicherry.
9. Jagdeesh C. Kalla; Vidyasagar Rao, K. and Pradeep Kumar presented a paper 'Economic viability of goat production system under intensive management', at International Goat Conference, 1991 at New Delhi.
10. Kulkarni, J.H. participated in a Course on Extension Management Facilitators Programme, organised by Coverdale Training Ltd, London, from June 3 to 21, 1991 at MANAGE, Hyderabad.
11. Mehra, U.R. and Jagannadham Challa presented a paper 'Rumen fermentation pattern in buffalo calves for urea - molasses mineral block in the diet' at the International Animal Nutrition Workers Conference for Asia and Pacific held at Bangalore from September 23 to 28, 1991.
12. Raghavendra Rao, V. participated in the group discussion on Management of Change in All India Group Improvement Programme from February 24 to 26, 1992 sponsored by ICAR at Directorate of Rice Research, Hyderabad.
13. Raman, K.V. and Balaguru, T. presented a paper 'Successful implementation of FSR/E in India: reliance on appropriate HRD efforts' at the FSR/E Workshop held at NAARM from November 25 to 28, 1991.
14. Raman, K.V., Samanta, R.K., Jagannadham Challa, and Shanmugam, S. participated in the Workshop on 'Appropriate Training Technologies for Bringing out Attitudinal changes', sponsored by the Andhra Pradesh Police Academy, Hyderabad on March 16, 1991. A paper on 'Attitudinal Changes in Agricultural Research Service Scientists in India' by Shanmugam, S. and Jagannadham Challa was circulated to the participants of the Workshop.
15. Rao, K.P.C. participated in the Fiftyfirst Conference of the Indian Society of Agricultural Economics held at APAU, Hyderabad from December 18-20, 1991.
16. Sabarathnam, V.E. attended the National Seminar on Agricultural Extension, organised by the Directorate of Extension, Ministry of Agriculture, New Delhi from February 25 to 26, 1992.

17. Samanta, R.K. presented a paper 'Appropriate Technology for sustainable Development in Agriculture' at National Seminar on Identification of Appropriate Farm Technologies for Rural Development and the Role of Marketing, Credit and other Institution including Government Agencies, held in Calcutta on September 28-29, 1991 by BCKV and WBCADC.
18. Samanta, R.K. attended the Workshop on 'Aquaculture Extension Research Needs' and presented a paper on 'Need of the hour is to change extension strategy for effective defusion and higher adoption of agricultural technology at Central Institute of Freshwater Aquaculture', Bhubaneswar, Orissa on June 20, 1991.
19. Samanta, R.K. attended a special Seminar of 'Participatory Approach to Extension Training' delivered by Dr Lautin Murs, Head, Agricultural Extension, International Agricultural Centre, University of Wageningen, the Netherlands at MANAGE, Hyderabad on October 11, 1991.
20. Shanmugam, S and Jagannadham Challa participated in two day National Consultation on ICAR Personnel Policies at NAARM on October 24-25, 1991.

Training

* Dr A. Gopalam, Associate Professor, underwent advanced training in Educational and Instructional Methodology at Michigan State University and Syracuse University, Syracuse in USA for a period of four-and-a-half months (March 22 - August 4, 1991) under the UNDP Project on Advanced Centre for Postgraduate Agricultural Research and Education. In addition, he participated in various courses, viz. i) Interactive video application in education, ii) Artificial intelligence and expert systems in education, iii) Programme on analysing the instructional problems and managing an instructional project, iv) Developing expertise in the production of computer aided self-instructional material, and v) Developing expertise in the production of a video tutorial during his stay in USA.

* Dr M.M. Anwer, Associate Professor, underwent advanced training in Educational Administration and Management at the Sloan School of Management of the Massachusetts Institute of Technology and Harvard Business School, Harvard University, Cambridge, USA for a period of four-and-a-half months (September, 9 1991- January 22, 1992) under the UNDP Project on Advanced Centre for Postgraduate Agricultural Research and Education. He attended various courses in the areas of Managerial Psychology, Individuals, Groups and Organisations, R.D. Process and Communication in organisation, the Management of Quality and Managing Centre.

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The Union Minister of State for Agricultural Research and Education, Mr K.C. Lenka, visited the Academy on August 18, 1991. Mr. Lenka held discussions with the Director, Dr K.V. Raman, and the faculty. He went round the physical facilities. Mr Lenka, flanked by Dr Raman (extreme right) and Dr Jagdeesh C. Kalla (extreme left), releasing the Annual Report 1990-91. Later, the minister addressing the 37th batch of Agricultural Research Service trainees said that agriculture holds the key for the nation's prosperity, it should be accorded the highest priority by the planners. Stating that agricultural scientists have great role to play in facing the future challenges on the food front, he exhorted the scientist-trainees to brace themselves for the task.

DISTINGUISHED VISITORS

A large number of distinguished personalities including national and international scientists visited the Academy during 1991-92. They include:

- ** Mr C. Srinivasa Sastry, former Agricultural Secretary and Member of the External Review Committee of the International Service for National Agricultural Research (ISNAR), the Netherlands, visited the Academy on June 26, 1991 and held discussions with the Director.
- ** A group of Ph.D. students from Andhra Pradesh Agricultural University taking a course on Agricultural Research Management visited the Academy on June 10, 1991. Dr. Devesh Kishore briefed them on the activities of the Academy with particular reference to the programmes on Agricultural Research Management.
- ** A USAID - Integrated Pest Management Study Team comprising Dr R.E. Ford, Dr A.C. Steinhauer and Dr C.R. Edwards, visited the Academy on April 25, 1991. The Team visited the various physical facilities and held discussions with the Director.
- ** A team of five officials from Zimbabwe, which attended the International Conference on Extension Strategy for Minimizing Risk in Rainfed Agriculture, visited the Academy on April 22, 1991. The team visited various facilities and held discussions with the Director.
- ** Dr Nicholas Haghes, Country Project Officer, Food and Agriculture Organisation visited the Academy on April 2, 1991, and held discussions with the Director. He also visited the Video Instruction and Production Laboratory and computer facilities created under the auspices of the FAO/UNDP.
- ** About 15 participants of the rank of Deputy Inspector General of Police, attending the course on 'Management of Training' at the National Police Academy visited NAARM on April 11, 1991. Dr Devesh Kishore addressed the group on "Institution Building".
- ** Mr Csaba Ipsits, Research Fellow of Research Institute for Irrigation Section for Rice Breeding and Production, Hungary, visited the Academy on July 2, 1991.
- ** Mr F.W. Lancaster, Graduate School of Library and Information Science, University of Illinois, visited the Academy on July 6, 1991

- ** Mr M.D. Abdus Siddique, Principal Training Officer, Bangladesh Agricultural Research Council, visited the Academy on July 8, 1991.**
- ** Prof. H.P.M. Gunasena, Dean, Faculty of Agriculture, University of Peradeniya, Peradeniya, Sri Lanka, visited the Academy on July 26, 1991.**
- ** Prof. H.M. Walter Herath, Department of Agricultural Biology, University of Peradeniya, Sri Lanka, visited the Academy on July 26, 1991.**
- ** Dr Kirti Singh, Vice-Chancellor, Himachal Pradesh Krishi Vidyalaya, Palampur, visited the Academy on September 5, 1991.**
- ** Dr (Mrs.) P. Sundaram, Dean, College of Home Science, Udaipur, visited the Academy on September, 5, 1991.**
- ** Mr Ore Bjerregaard, Deputy Resident Representative, United Nations Development Project, New Delhi, visited the Academy on September 17, 1991.**
- ** Dr Har Swarup Singh, Lt. Governor, Pondicherry, visited the Academy on September 9, 1991**
- ** Dr. N.V. Koteswara Rao, Director, National Informatics Centre, Hyderabad, visited the Academy on September 24, 1991.**
- ** Mr William C. Nickel, Senior Agricultural Economist, World Bank, Washington, D.C. USA, visited the Academy on September 25, 1991.**
- ** Dr Byron Mook, ISNAR, visited the Academy on October 4, 1991. He held discussions with the Director.**
- ** Dr J.B. Adégbeye, Alh. Mohammed Nagenu and Mrs Fofah, from Federal Agricultural Coordinating Unit, Government of Nigeria, visited the Academy on October 23, 1991.**
- ** Prof. Ndanusa B. Mijindadi, Head, Federal Agricultural Coordinating Unit, Ibadab, Nigeria visited the Academy on November 18, 1991**
- ** Mr Gerry Roberts, Extension Officer, Arid Zone Institute, Department of Primary Industries, Queensland Government, Australia, visited the Academy on December 13, 1991.**
- ** Dr Sewaram, Senior Scientist, Project Directorate on Cropping System & Research, Modipuram, visited the Academy on December 6, 1991 and held discussions with Dr T. Balaguru.**

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The US-AID Integrated Pest Management study team visited the Academy on April 25, 1991.



Mr. R.C. Kapila, Director General, ICAR, (third from left) visited the Academy on May 10, 1991, and held discussions with the Director and faculty members. He was accompanied by Dr. R.S. Paroda, Deputy Director General (Crops), ICAR (Extreme left) and Dr. G.C. Srivastava, Secretary, ICAR (Second from right). Also seen in the picture are Dr. K.V. Raman, Director and Dr. Jagdeesh C. Kalla, Joint Director.

- ** A batch of 53 B.Sc.(Ag.) students from the Kerala Agricultural University visited the Academy on December 11, 1991.
- ** A three member delegation from People's Republic of China comprising Dr Chen Wanjin, Vice President, Dr Zhang Lijian, Deputy Director of International Exchange Division and Mr Wang Zhongqiu, Project Officer, Chinese Academy of Agricultural Sciences, Beijing visited the Academy on March 3, 1992. They held discussions with the Director-in-charge, Dr Jagdeesh C. Kalla and with the Unit Heads. They visited the Video Instruction and Production Laboratory, Self-aided Instruction Laboratory, Computer Laboratory and the Library. Dr V. Raghavendra Rao coordinated the visit.
- ** Two batches of 70 each, undergraduate students from the University of Agricultural Sciences, Dharwad, visited the Academy on February 18 and 20, 1992, respectively.
- ** About 20 police officers in the rank of Deputy Inspector General of Police undergoing management training at the National Police Academy, visited the Academy on February 12, 1992. Dr Devesh Kishore interacted with the participants on important aspects of institutional building.
- ** Mr. S.I. Zakhariev, FAO representative, visited the Academy on March 23, 1992. He held discussions with the Director-in-charge and Unit Heads on the UNDP programmes.

NAARM ALUMINI ASSOCIATION

The total membership of the Association stands at 473 as on March 31, 1991.

Following are the office-bearers of the Association for 1991-92:

PATRON	:	Director General, ICAR
PRESIDENT	:	Dr K.V. Raman, Director, NAARM
GENERAL SECRETARY	:	Mr R. Chittiraichelvan, Sr. Scientist, NAARM
VICE-PRESIDENTS (2)		
Local	:	Dr Vilas A. Topani, Scientist NRCS, Hyderabad.
All India	:	Dr A.K. Saxena, Scientist CIAE, Bhopal
JOINT SECRETARIES (5)		
Eastern Region	:	Dr K.K. Sharma, Scientist ILRI, Ranchi
Northern Region	:	Dr Sanjeev Gupta, Scientist IGFRI, Jhansi
Western Region	:	Mr A. Chaturvedi, Scientist NSSB & LUP, Nagpur
Central Region	:	Mr U.R. Badegaonkar, Scientist, CIAE, Bhopal
Treasurer	:	Mr C.P. Mohan Kumar Finance & Accounts Officer NAARM, Hyderabad

IMPORTANT COMMITTEES

Management Committee

The Academy is guided in planning, organizing and conducting various types of training programmes and research studies by the Management Committee of NAARM. The Management Committee approves the programmes to be undertaken. The Academy ensures their implementation, both in terms of physical targets and schedules, and provide broad guidelines for future programmes. The Management Committee also helps in formulation and consideration of proposals for Five Year Plan and Annual Plan and takes up periodical review of progress of development schemes.

The Management Committee, besides assisting Director of the academy in the discharge of his functions, focuses his attention on the research and other programmes of the academy and ensures their implementation. While examining these programmes periodically, it pinpoints bottlenecks, if any, and suggests remedial measures.

The 18th and 19th meetings of the Management Committee of the Academy were held on August 17, 1991 and January 29, 1992, respectively. The composition of the Management Committee of NAARM is as follows:

- | | | |
|----|---|----------|
| 1. | Dr K.V. Raman
Director, NAARM
Rajendranagar
HYDERABAD - 500 030 | Chairman |
| 2. | Dr Jagdeesh C. Kalla
Joint Director, NAARM
Rajendranagar
HYDERABAD - 500 030 | Member |
| 3. | Dr N.K. Bhattacharya, Director
Central Institute for
Research on Goats
P.O. FARAH - 281 122
MATHURA (U.P) | Member |
| 4. | Dr V.R.P. Sinha, Director
Central Institute of
Fisheries Education
Versova, BOMBAY - 400 058 | Member |

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| 5. | Dr N.T. Singh, Director
Central Soil Salinity
Research Institute
KARNAL - 132 001 (Haryana) | Member |
| 6. | Mr V.S. Sampath
The Commissioner &
Director of Agriculture
HYDERABAD | Member |
| 7. | The Director of Animal Husbandry
Govt. of Uttar Pradesh
LUCKNOW (U.P) | Member |
| 8. | Dr M.V. Rao
Vice-Chancellor
Andhra Pradesh Agricultural University
Rajendranagar
HYDERABAD - 500 030 | Member |
| 9. | Mr N.B. Reddy
Post: AURAD
(Sirsi), Dist. Bidar
(KARNATAKA) | Member |
| 10. | Dr D. Bap Reddy
33/4 RT, Barkatpura
HYDERABAD - 500 027 | Member |
| 11. | Dr R. Nagarcenkar
Deputy Director General (Edu.)
Indian Council of Agric. Research
Krishi Bhavan
NEW DELHI - 110 001 | Member |
| 12. | Mr D.K. Srivastava
Director (Finance)
Indian Council of Agric. Research
Krishi Bhavan
NEW DELHI - 110 001 | Member |
| 13. | Mr K.K. Bajpai
Chief Adm. Officer
NAARM, Rajendranagar
HYDERABAD - 500 030 | Member-Secretary |

Official Language Implementation Committee

The Official Language Implementation Committee of the Academy provided periodical suggestions for implementation of Official Language Policy, and Rules and Regulations of Govt. of India that are issued from time to time. The Committee met on August 23, 1991; February 3, 1992; and March 30, 1992 and discussed several issues relating to the implementation of Hindi language. The following are the members:

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|-----|--|---------------------------------|
| 1. | Dr Devesh Kishore | Chairman
Principal Scientist |
| 2. | Dr K. Venkateswarlu
Sr Scientist | Member |
| 3. | Dr A. Gopalam
Sr Scientist | Member |
| 4. | Dr Jagannatham Challa
Sr Scientist | Member |
| 5. | Mr R.B. Gaddagimath
Librarian | Member |
| 6. | Mohd. Habeebuddin
Chief Admn. Officer i/c | Member |
| 7. | Mr P. Mohan Kumar
Finance & Accounts Officer | Member |
| 8. | Mr M. Srinath Reddy (up to 29.08.91)
Public Relations Officer | Member |
| | Mr Vijay Kumar (from 18.01.92)
Assistant Editor | Member |
| 9. | Mr R.V.V.S. Prakasa Rao
Editor | Member |
| 10. | Mr P.P. Brahmaji
Assistant | Member |
| 11. | Mr P. Swamy
SSG-I | Member |
| 12. | Mr D. Venkateswarlu
Technical Officer (Hindi) | Member Secretary |

PERSONNEL

K.V. Raman
Jagdeesh C. Kalla

Director (up to 31.01.92)
Director-in-charge (w.e.f. 01.02.92)

Director's Personal Cell

L. Jhansi Lakshmi
T.V. Rama Das
P.Swamy

Sr. Stenographer
Jr. Stenographer
SSG-I

AGRICULTURAL RESEARCH SYSTEMS AND POLICIES UNIT

Jagdeesh C. Kalla
P. Kumar
VE. Sabarathnam
T. Balaguru
K.P.C. Rao
K. Vidyasagar Rao
Sarada Samanta
A. Vijaya Lakshmi

Joint Director (up to 31.01.92)
Principal Scientist
Sr. Scientist
Sr. Scientist
Sr. Scientist
Sr. Scientist (on deputation)
Sr. Stenographer
Jr. Stenographer

Computers Section

U. Muralidhar
M.J. Pradeep Kumar
K. Venkateswara Kumar
C.V.V.S.S.R.L. Phani Raj

Programmer (Computer)
Computer Operator
Statistical Assistant
Jr. Clerk

EDUCATIONAL SYSTEMS AND TECHNOLOGY UNIT

Devesh Kishore
R. Sudarsanam
A. Gopalam
K.M. Reddy
Rama Rao
J.H. Kulkarni
K.R. Prabhakar
N.R. Nageswara Rao
Ch. Janardhan Rao
P. Anantha Reddy
N. Naresh Kumar
M. Venkatesh

Principal Scientist
Principal Scientist (up to 11.06.91)
Sr. Scientist
Sr. Scientist
Sr. Scientist (from April 15, 1991)
Sr. Scientist (up to 29.02.92)
Programme Officer (CCTV)
Technical Officer
Video Cameraman
Technician (Instruments)
VCR-cum-Tele Cine Operator
Stenographer

Hindi Section

D. Venkateswarlu
J. Renuka
S. Pradeep Singh
P.R.R. Jyothi

Technical Officer (Hindi)
Hindi Assistant
Hindi Translator
Jr. Clerk

PROJECT MANAGEMENT UNIT

V. Raghavendra Rao
K. Venkateswarlu
S. N. Saha
P. Manikandan
T. Vanisri

Principal Scientist
Sr. Scientist
Sr. Scientist
Sr. Scientist
Jr. Stenographer

TRANSFER OF TECHNOLOGY SYSTEMS AND POLICIES UNIT

R.K. Samanta
R.C. Chelvan
A.U. Gadewar
L. Venkateswarlu
Bansidhar Nayak
D. Rajagopala Rao
S. Sunder Raj
P. Anand Kumar

Sr. Scientist
Sr. Scientist
Sr. Scientist
Photographer
Projectionist
PAE Operator
Media operator
Jr. Stenographer

HUMAN RESOURCES DEVELOPMENT UNIT

M.M. Anwer
Jagannadham Challa
S. Shanmugam
Y. Anuradha

Sr. Scientist
Sr. Scientist
Sr. Scientist
Jr. Stenographer

INFORMATION AND DOCUMENTATION UNIT

R.B. Gaddagimath
P.V. Nirmala
Y.V. Siva Prasad
T. Shanmuganathan
A. Mercy
D. Dhanalakshmi
Venkatesham
M. Narsing Rao

Librarian
Cataloguer
Documentation Assistant
Library Assistant
Jr. Stenographer
Jr. Clerk
Xerox Machine Operator
SSG-I

PRODUCTION AND PUBLICATION SECTION

R.V.V.S. Prakasa Rao
N. Narasimha Reddy
A. Murali Mohan Rao
P. Namdev
K. Anil Kumar
P. Vijender Reddy
Y. Vijay Kumar
M. Shekhar Reddy
S. Swamy
Prashant Gaikwad
C. Chandramouli

Editor
Off-set Machine Operator
Artist
Graphic-cum Visualiser
Reprographic Assistant
Museum Assistant
Assistant Editor
Darkroom Assistant
Asst. Gestt. Operator
Binder
SSG-I

FARM SECTION

V. Murali
K. Rangasamy
M.A. Basith
B. Veeraiah
Ahire Laxman Maharu
K. Shivaiah
K. Prabhudas
M. Padmaiah
K. Veeranarasaiah
P. Eswari
P. Balraj
M. Yadaiah
K. Satyanarayana
N. Sukunamma
T. Jangamma
S. Jangaiah
B. Bharatamma
J. Chandraiah
G. Pentaiah
R. Sattaiah
G. Anasuya
K. Hemlal

Farm Superintendent
Field Assistant
Lab Technician
Farm Assistant
Farm Assistant
Technician (Electrical)
Sr. Clerk
Tractor Driver
Carpenter-cum-Painter
Power Tiller Driver
SSG-I
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HOSTEL SERVICES

Zameer Ahmed
Sham Bahadur
D. Sangamma
C. J. Samuel

Manager (Hostel Services)
Catering-in-charge
Cook
Jr. Clerk

G.V. Bikshapathi
 B. Santamma
 B. Subba Rao
 Shakunthla
 G. Dasaradha
 A. Yadaiah
 M. Krishnaiah
 C. Anjaiah
 K. Kalavathi
 M. Yadamma
 I. Bharatamma

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ADMINISTRATION

(a) Office:

S. Rengan
 K.K. Bajpai
 Mohd. Habeebuddin
 M. Srinath Reddy
 K. Janakiramaiah
 T. Ashok Kumar
 Y. Sankara Rao
 P.P. Brahmaji
 K.S. Sharma
 N. Raghunath
 P.G. Kohad
 P. Neelakantam
 W. Srinivasa Bhat
 M. Narasimha Rao
 T. Srinivas
 P.P. Sastry
 M. Dinesh
 K. Swarajya Lakshmi
 G. Raj Reddy
 B. Padma Saroja
 P. Srinivasu
 K. Srinivasa Rao
 P.D. Jessie Ecclicia
 M. Sridhar
 C. Bikshapathi
 M. Sham Rao

Chief Adm. Officer (up to 31.10.91)
 Chief Adm. Officer (from 19.12.91)
 Asst. Adm. Officer
 Public Relations Officer (up to 29.08.91)
 Superintendent
 Assistant
 Assistant
 Assistant
 Assistant
 Stenographer
 Assistant
 Assistant
 Sr. Clerk (on deputation)
 Sr. Clerk
 Sr. Clerk
 Jr. Stenographer
 Jr. Clerk
 Jr. Lib. Asst
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 PMBX Operator-cum-Receptionist
 Jr. Clerk
 SSG-I
 SSG-I

(b) Vehicles:

K. Sattaiah	Driver
S.K. Peera	Driver
M.M. Mohan Rao	Vehicle Mechanic
Mohd. Naseer Khan	Driver
G. Muthyalu	Driver
U. Venkataratnam	Driver
N. Ashok	Driver
B. Narayana	SSG-II

(c) Security:

B. Ch. Satyanarayana	S.O.-cum-Caretaker
N. Prabhakara Rao	Plumber
G. Mani Bai	SSG-II
B.H. Dharma Raj	SSG-II
Phool Kumar	SSG-I
Khalid	SSG-I
M. Ashok	SSG-I
L. Satyanarayana	SSG-I
M. Eswaramma	SSG-I

(d) Electricals:

M.V.V.S. Ravindra Babu	Electrical Overseer (up to 06.03.92)
L. Ramesh	Electrician-cum-Plumber
M.K. Sonkusare	Electrician
M. Mohan Rao	Technician (Electrical)
M.K. Shamsuddin	Technician (Electrical)
B.S.N. Murthy	Technician (Electrical)
K. Obulpathi	Technician (Electrical)
B. Satyanarayana	Technician (Electrical)
A. Nageswara Rao	Lineman
D.R. Sahadev Rao	Pump Driver

AUDIT & ACCOUNTS SECTION

C.P. Mohan Kumar	Finance & Accounts Officer
C. Bagaiah	Assistant
M. Girija Kumari	Superintendent A&A
S. Balakamesh	Jr. Clerk
R. Chandra Babu	Jr. Clerk

Resignations

The following employees of the Academy were relieved from the service to take up new assignment at other organisations during the period:

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|------------------------------|--------------------------|
| 1. Mr M.V.V.S. Ravindra Babu | Electrical Overseer |
| 2. Mr M. Srinath Reddy | Public Relations Officer |
| 3. Mr T. Pardhasarathi | Jr. Clerk |

Promotions

The following members of the staff were promoted on recommendation by the Assessment Committee/Departmental Promotion Committee during the period:

S.No.	Name of the Employee	Designation	Promoted As
1.	Mr T. Ashok Kumar	Assistant	Superintendent
2.	Mr M. Dinesh	Jr. Clerk	Sr. Clerk
3.	Mr P. Neelakantam	Sr. Clerk	Assistant

On adhoc basis

4.	Ms. Girija Kumari	Jr. Stenographer	Superintendent
5.	Ms. Jhansi Laxmi	Stenographer	Sr. Stenographer

Transfers

The following transfers were made during the period (NAARM to other ICAR Institutes and viceversa):

S.No	Name of the Employee	Designation	From	To
1.	Dr J. H. Kulkarni	Sr. Scientist	NRCG, Junagadh.	NAARM
			NAARM	UAS, Dharwad
2.	Dr Rama Rao	Sr. Scientist	CTRL, Bombay	NAARM
3.	Mr K.K. Bajpai	Chief Adm. Officer	IVRI, Izatnagar	NAARM

Retirements



Raman retires

Dr K.V. Raman, Director of the Academy, retired on superannuation on January 31, 1992 after a distinguished service of eight years at NAARM.

Dr Raman, who professes the learner-centred approach, has developed a unique organizational structure for the Academy in this endeavour. He modified the foundation course training programme for the Agricultural Research Service scientists, shifting the focus from the teacher to the learner. Innovative and need-based training programmes for the senior scientists of ICAR and faculty members of Agricultural Universities were also developed under his expert guidance.

Besides making an indelible mark on the training programmes, Dr Raman steered the FAO-UNESCO-UNDP supported Advanced Centre for Agricultural Education and Research Management at the Academy. Under this programme, infrastructural facilities like Video Instruction and Production Laboratory (VIPL); Self-Aided Instruction Laboratory (SAIL) for individualized and programmed learning; Audio-Visual Laboratory; a Computer Laboratory with multi-terminal facilities were established. A library to make teaching-learning process more effective was also added. He was mainly instrumental in developing the NAARM Faculty by sending them for advanced professional training to various leading management institutes in USA.

Dr S. Rengan, Chief Administrative Officer, retired from NAARM/ICAR services on superannuation on 31.10.91.

Deputations

The following members of the staff were on deputation:

Dr K. Vidyasagar Rao	Sr Scientist	Sr. Statistician (w.e.f. 09.10.91) ICRISAT, Hyderabad
Mr W. Srinivasa Bhat	Sr. Clerk	Assistant (w.e.f. 03.10.91) NRCS, Hyderabad

